

Epistemic AI Agents for Exploring Chemical Design and Synthesis Spaces

Talk at Technion
Department of Materials Science and Engineering
1.1.2026

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Department Information Systems
Faculty of Computer and Information Science
University of Haifa



Outline

1. Personas of Agentic AI

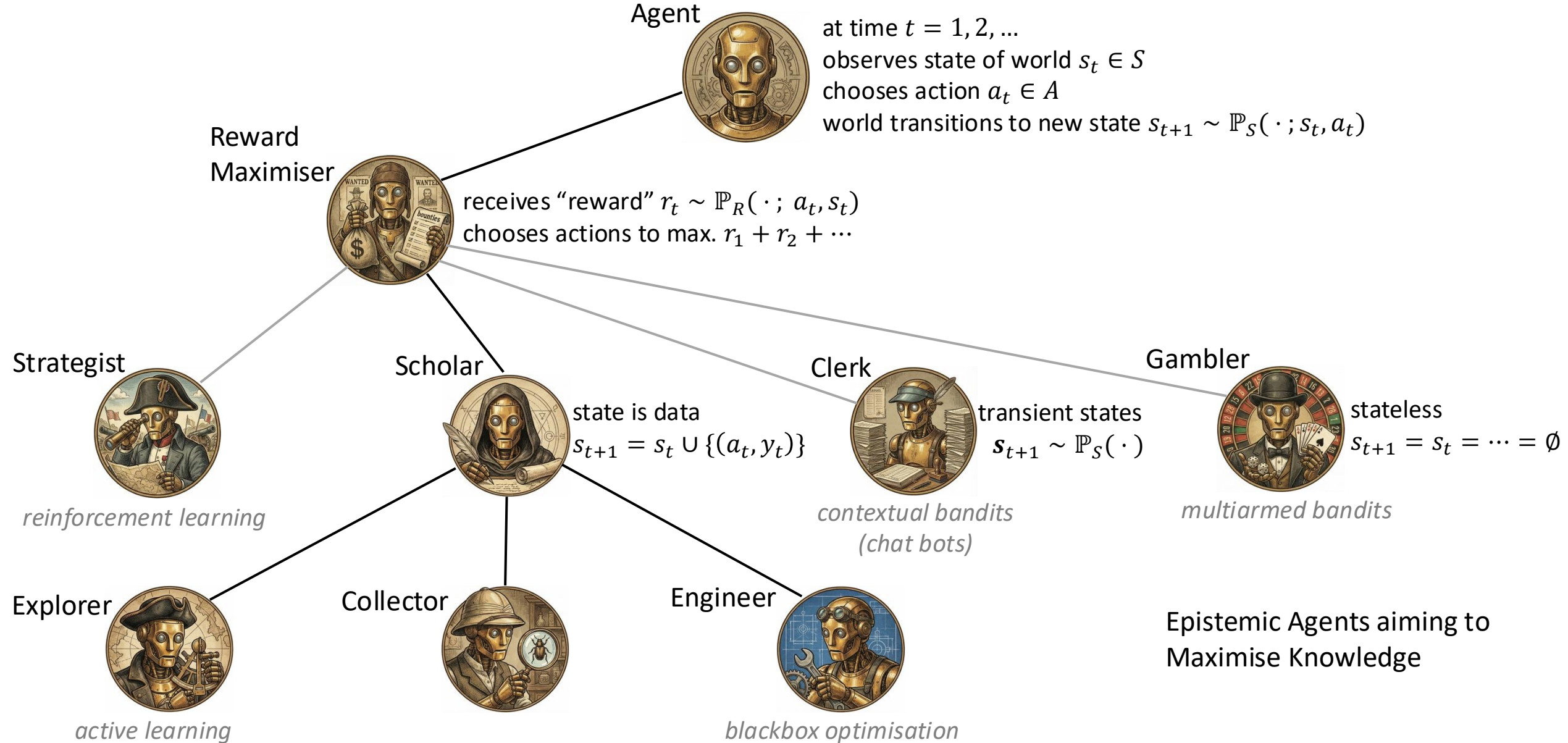
2. *Explorer*: mapping polymerisation-induced self-assembly phases

3. *Engineer*: optimising double perovskite hardness

4. *Collector*: discovering novel metal-organic frameworks

5. Problems and directions

AI Agents come in different “Personas”



Epistemic Agent: State = Data acquired by Actions

World behaviour captured in
conditional distributions

$$y \mid x$$

of

- target variable y given
- input variable $x \in X$

Actions $A = X^k$ allow agent to accrue
data $s_0 \subset s_1 \subset s_2 \subset \dots$

action
 a_2

x_1	y_1
...	...
x_k	y_k
x_{k+1}	y_{k+1}
...	...
x_{2k}	y_{2k}
...	...
x_{tk}	y_{tk}

state s_t

Agent chooses actions based on
response model and uncertainty

$$y \mid x, \theta$$

$$\theta \mid s$$

as usual, to maximise rewards...

Explorer



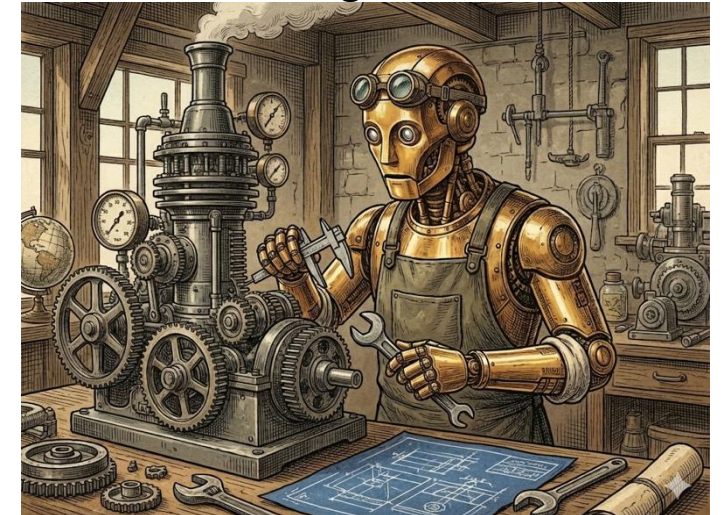
$$r_t = H(\theta \mid s_{t-1}) - H(\theta \mid s_t)$$

Collector



$$r_t = \#\{(z, y) \in Z \times Y_+ : (z, v, y) \in s_t \setminus s_{t-1}\}$$

Engineer



$$r_t = \max\{y : (x, y) \in s_t\} - \max\{y : (x, y) \in s_{t-1}\}$$

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Mapping the “Morphological Wonderland” of Polymerisation-Induced Self Assemblies



Yiwen
Lu



Dilek
Yalcin



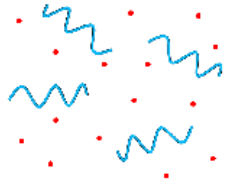
Lewis
Blackman



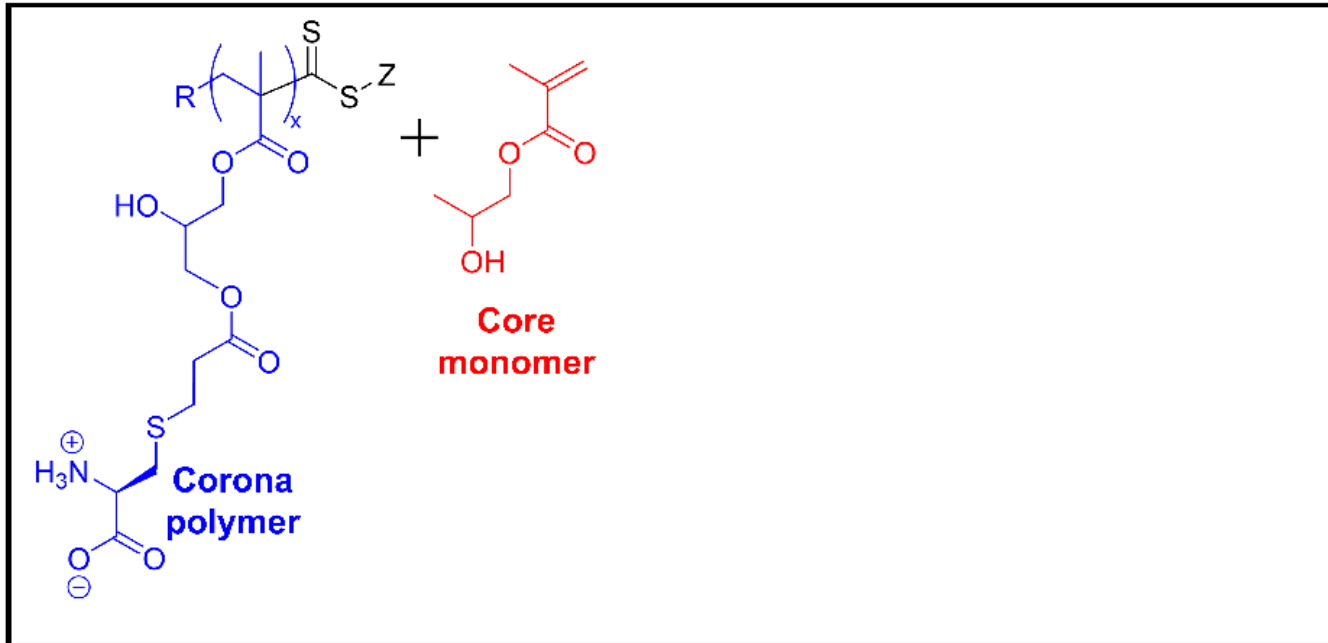
Goal: reduce the number of experiments required to obtain accurate phase diagrams of self assemblies of block copolymers

[Y Lu, D Yalcin, P Pigram, L Blackman, and M Boley. "Interpretable machine learning models for phase prediction in polymerization-induced self-assembly." *Journal of Chem. Inf. and Modeling* 63, no. 11 (2023): 3288-3306.]

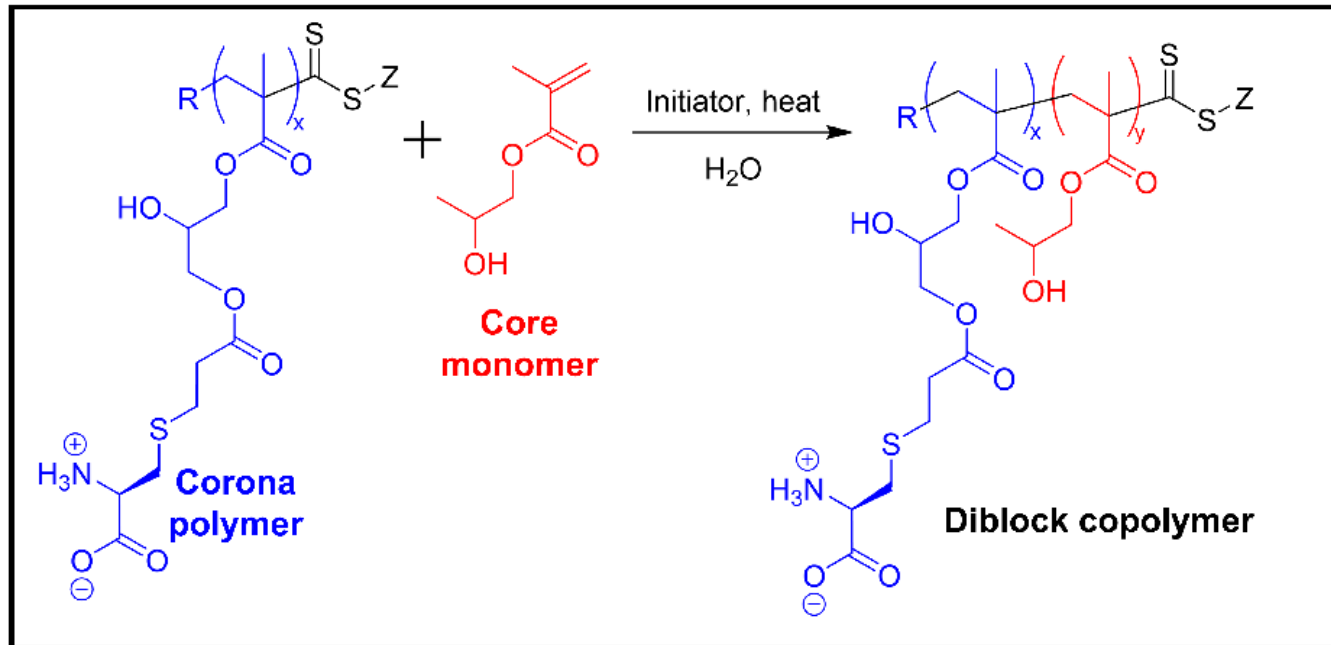
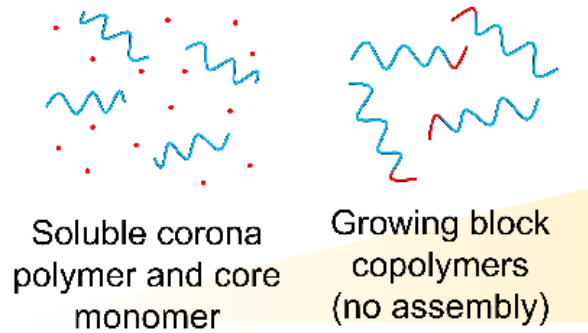
Polymer-Induced Self-Assemblies



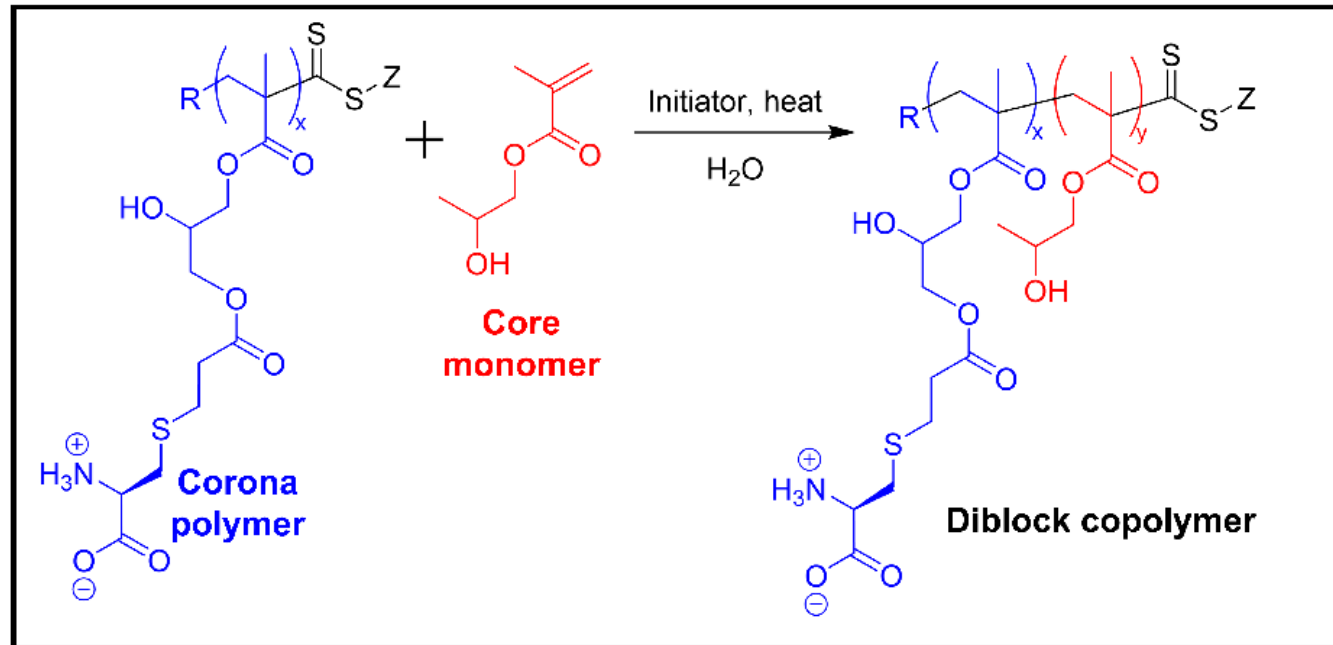
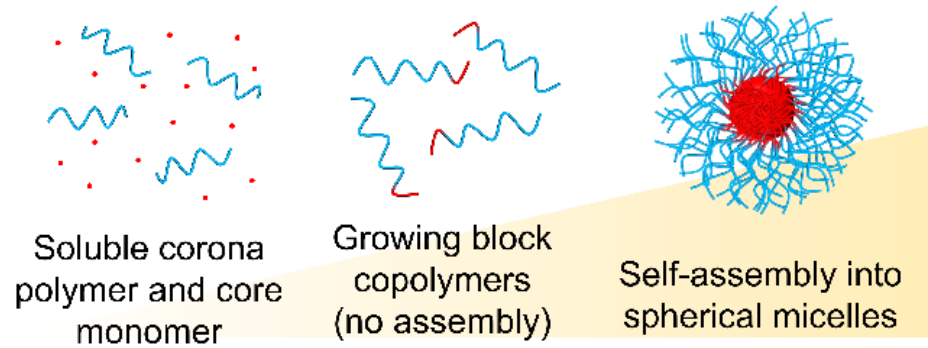
Soluble corona polymer and core monomer



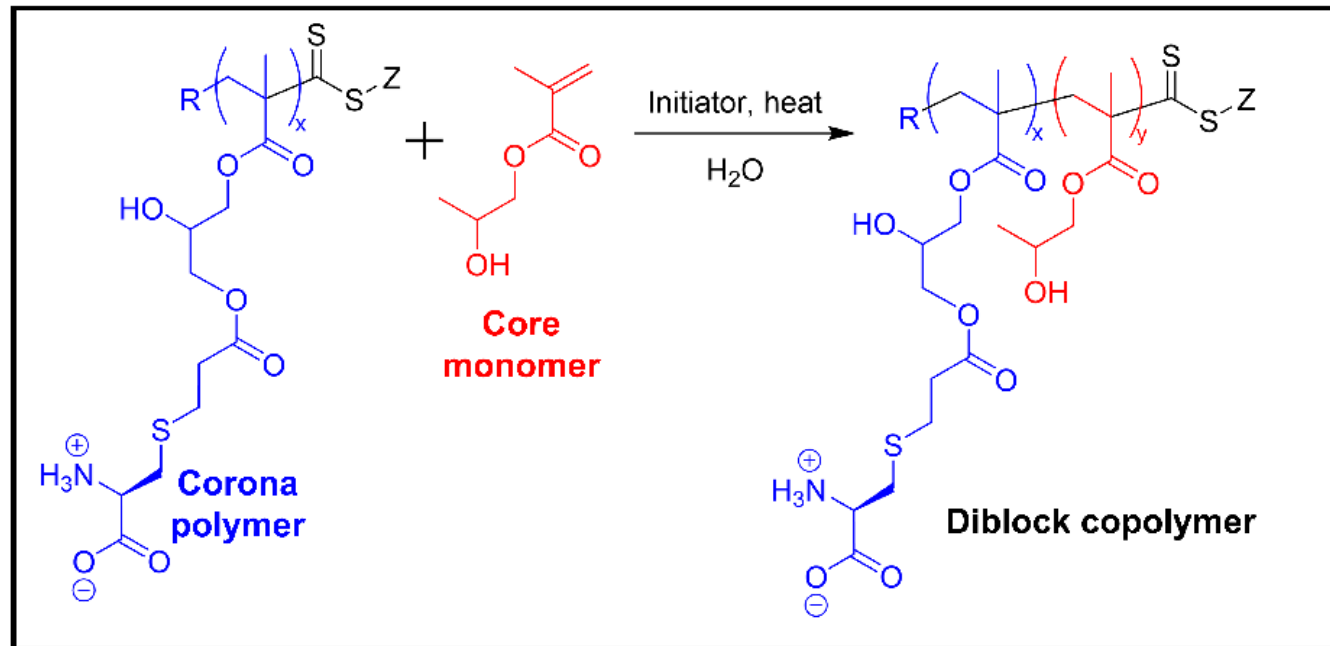
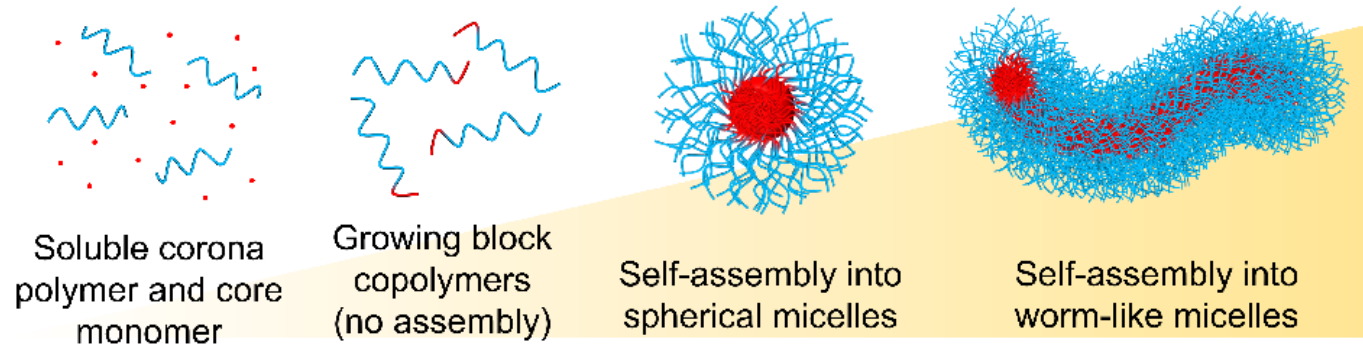
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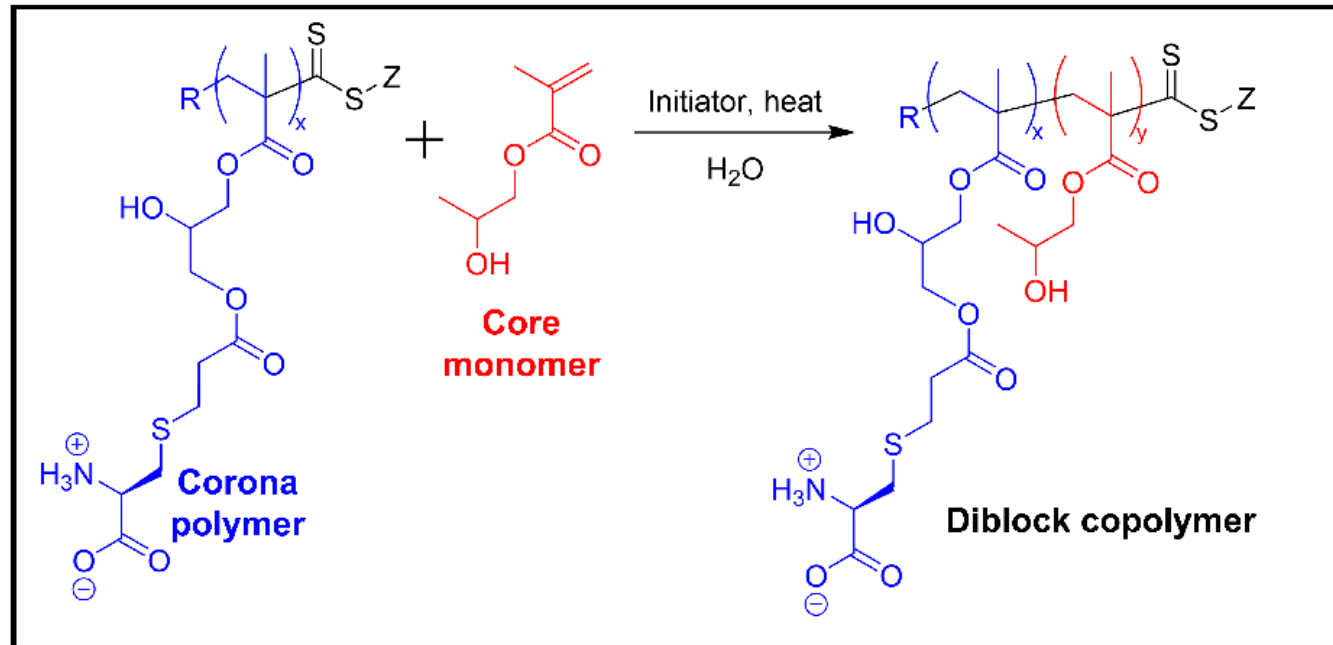
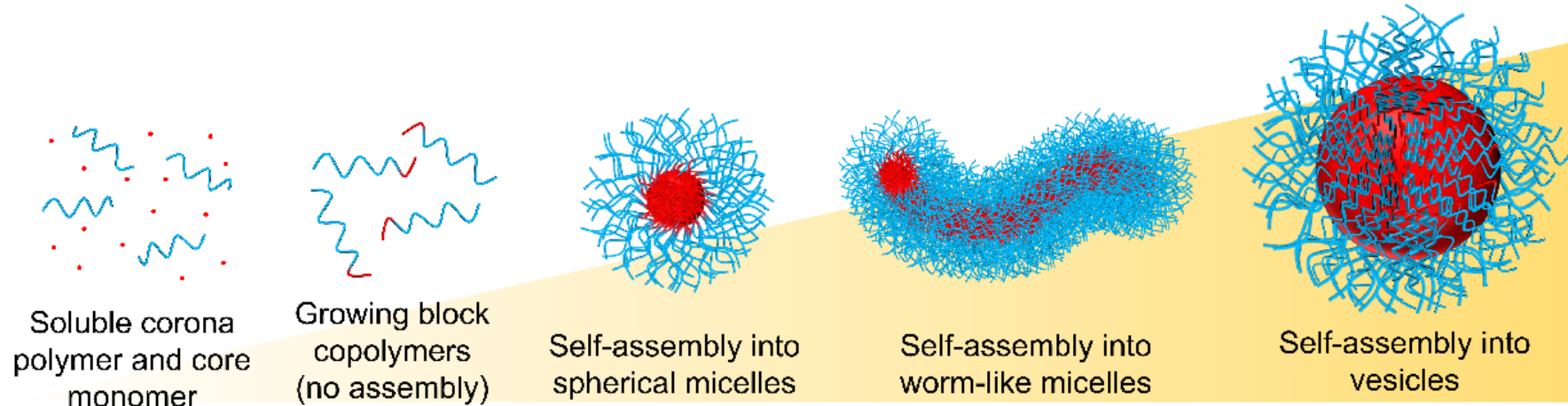
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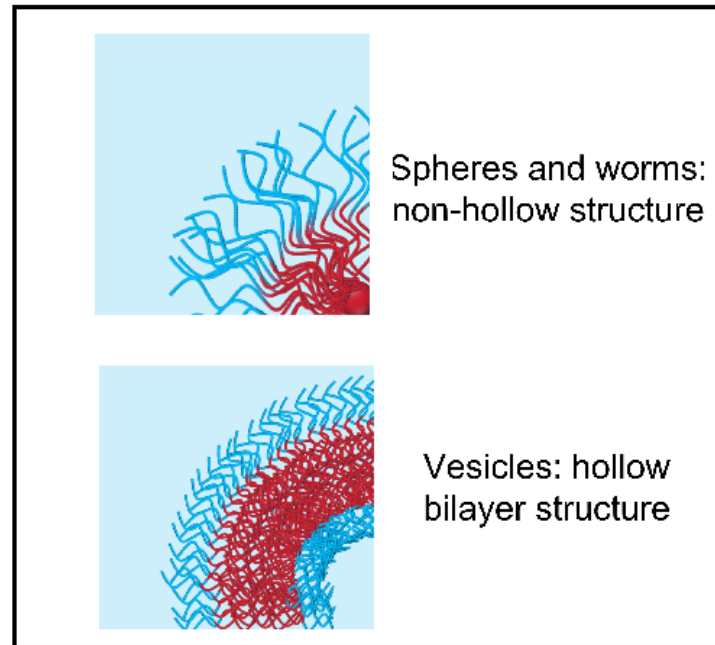
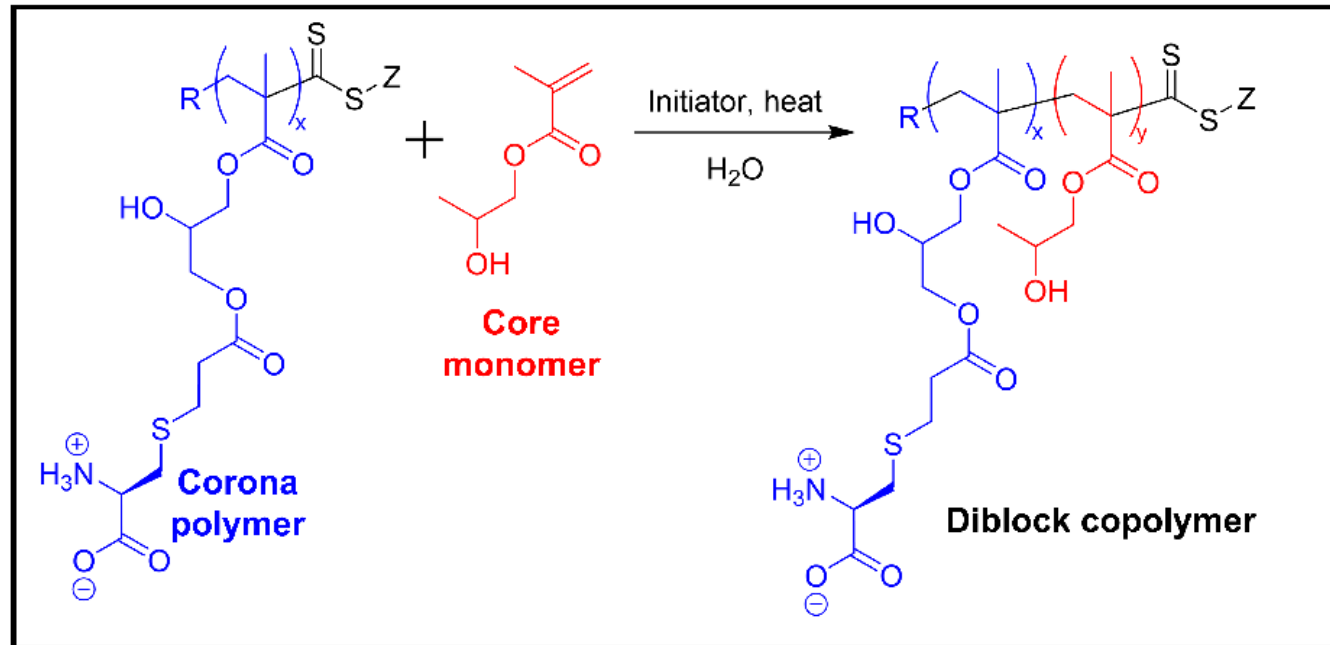
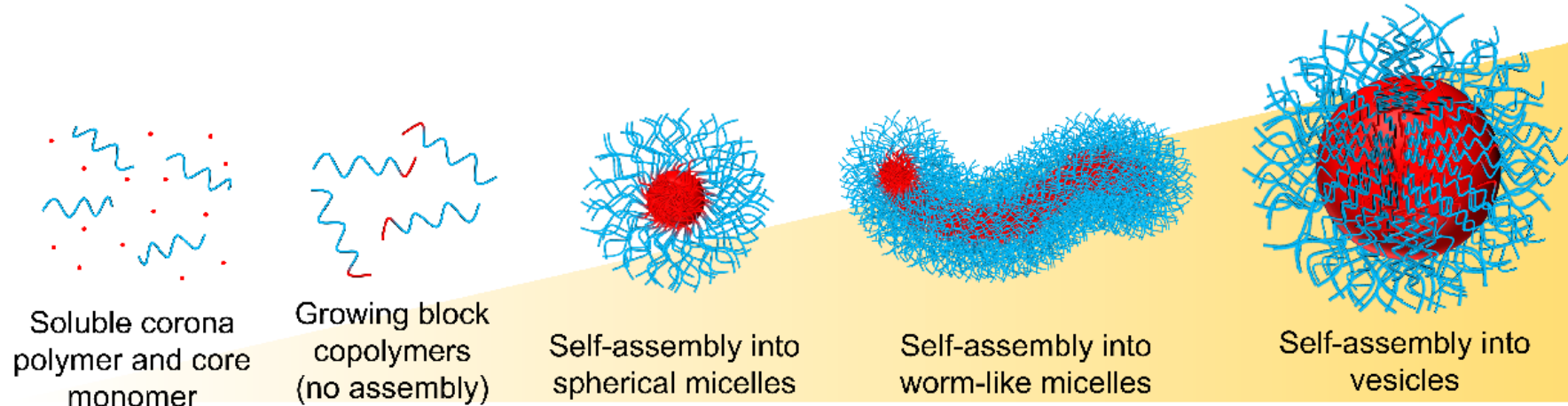
Polymer-Induced Self-Assemblies



Polymer-Induced Self-Assemblies



Polymer-Induced Self-Assemblies



Explorers Minimise Uncertainty about World



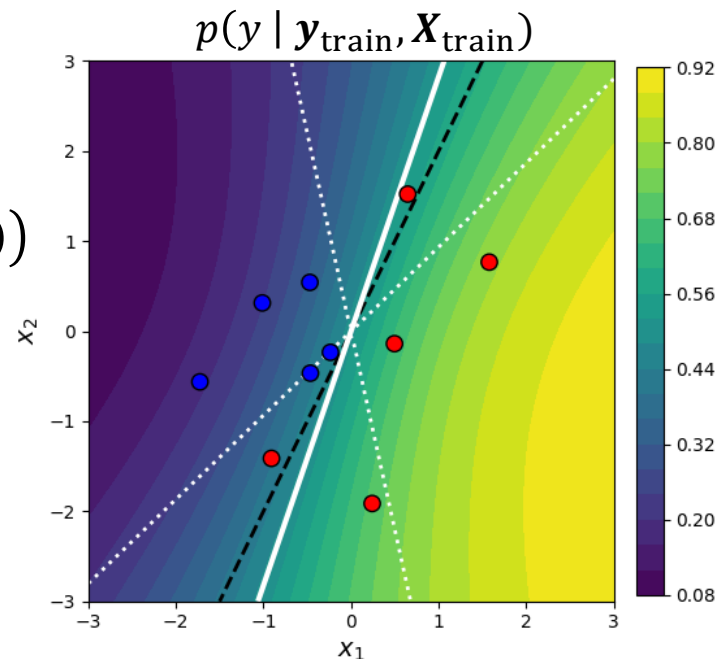
Ideas

- Model parameter **uncertainty** $\theta \mid s$
- Information **entropy** $H(\theta \mid s)$ captures how much left to learn in state s
- **Reward**: information gain $H(\theta \mid s_{t-1}) - H(\theta \mid s_t)$
- Data **utility**: expected information gain
$$u(x) = H(\theta \mid s) - \mathbb{E}(H(\theta \mid s \cup \{(x, y)\}))$$
$$= H(y \mid x, s) - \mathbb{E}(H(y \mid x, \theta))$$

Toy example: 2D Bayesian logistic regression

Model

$$\theta \sim N_2(0, I)$$
$$y \mid \theta \sim \text{Bern}(\sigma(x^T \theta))$$



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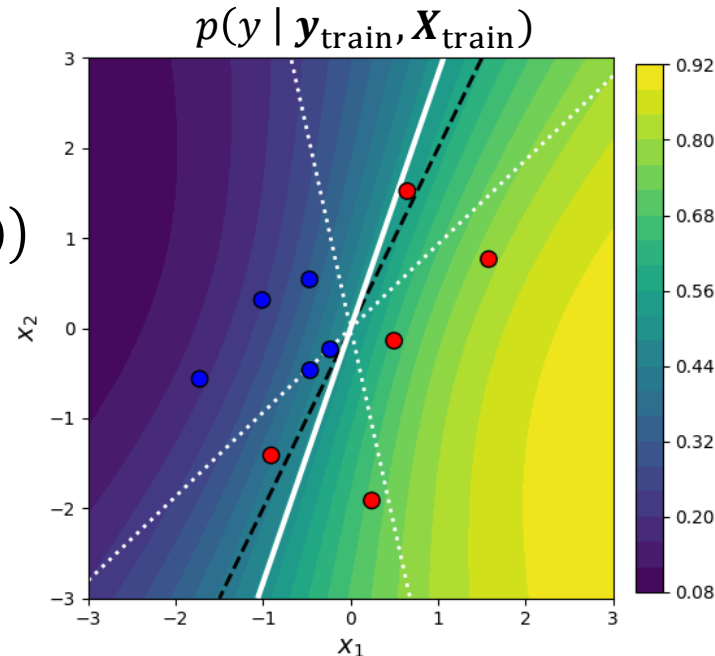
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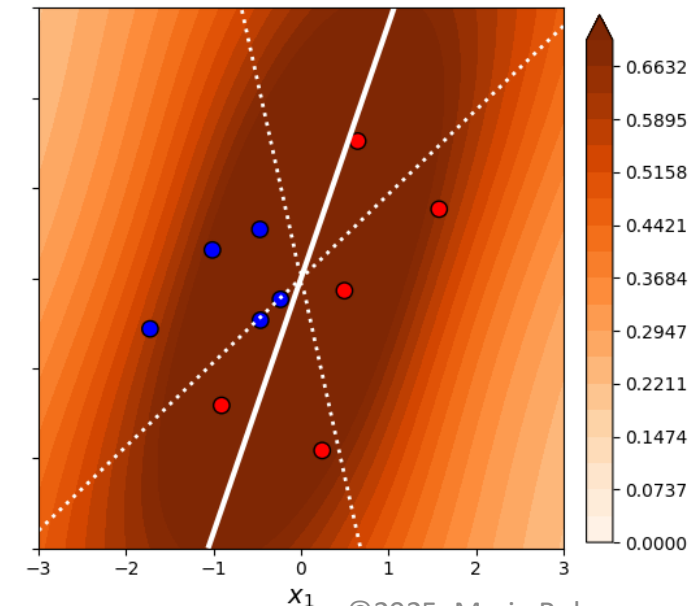
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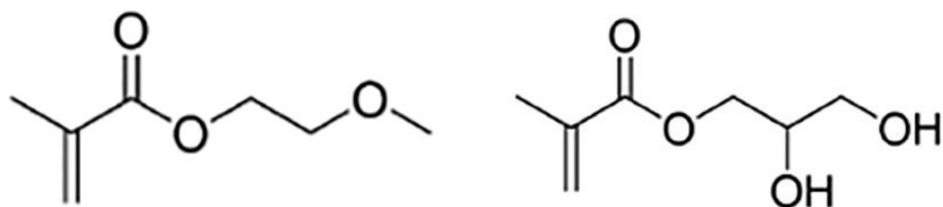
epistemic uncertainty

aleatoric “intrinsic” uncertainty



Morphological Outcome Models for PISA

Core and Corona Monomers



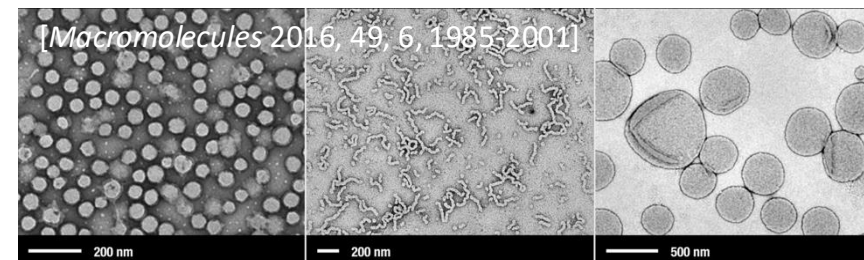
- molar mass
- molecular volume
- polar surface area
- sum of atomic polarizabilities
- log octanol/water partition coefficient (computed)

Conditions



- temperature
- degree of polymerization (core and corona)
- concentrations
- solvent pH
- ...

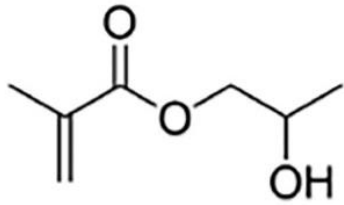
Outcomes



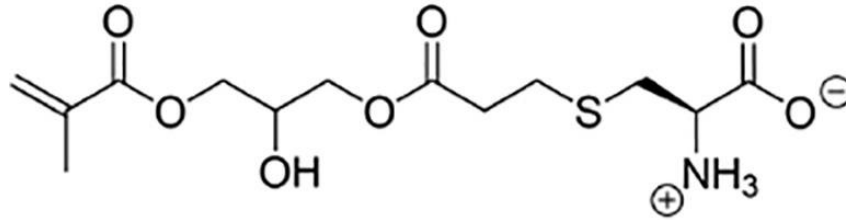
- spheres
- worms
- vesicles
- ...

Rapid Phase Diagrams for New Monomer Combinations

Core
HPMA



Corona
CysMA



Conditions

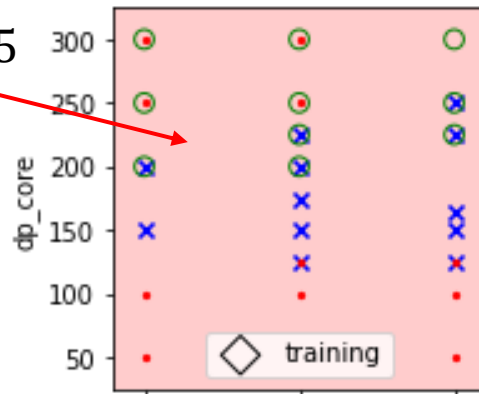
DP corona: 31

Temp: 70

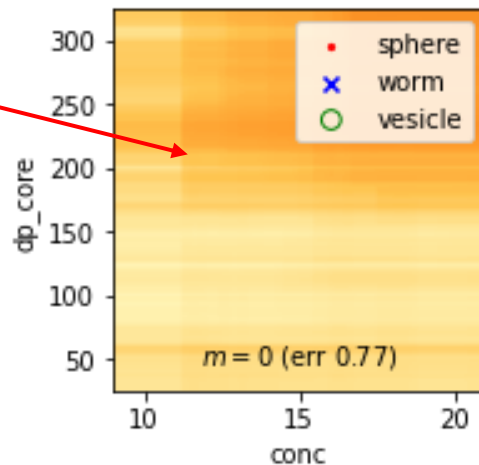
pH: 7

$$p(y_l | x) > 0.5$$

initially true
only for
“sphere”
everywhere



$$u(x)$$



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Double Perovskite Optimisation as Test Bed for AI-Tools Supporting Rational Materials Design



Simon
Teshuva



Phuc "Felix"
Luong



Daniel
Schmidt



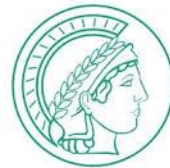
Lucas
Foppa



Matthias
Scheffler

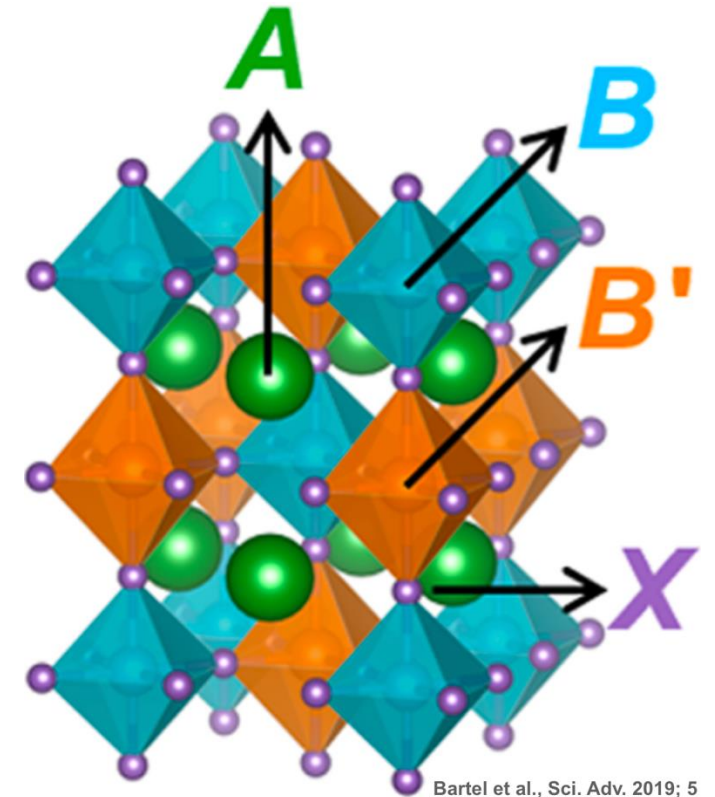


MONASH
University



MAX-PLANCK-GESELLSCHAFT
Fritz-Haber-Institut

Goal: development and evaluation of AI-tools for materials property optimization using the example of double perovskites



[M. Boley *et al.*, "From prediction to action: Critical role of performance estimation for machine-learning-driven materials discovery," in S. Bauer *et al.*, *Modelling Simul. Mater. Sci. Eng.* 32, 063301 (2024)]

Engineers Search for Optimal Design



Components

- design space $X = \{x_1, x_2, \dots\}$
- performance $y(x) \in \mathbb{R}$

Reward: improvement

$$r_t = y_t^* - y_{t-1}^*$$

$$y_t^* = \max\{y : (x, y) \in s_t\}$$

Acquisition: expected improvement

$$u(x; s) = \mathbb{E}(\max\{y - y_s^*, 0\})$$

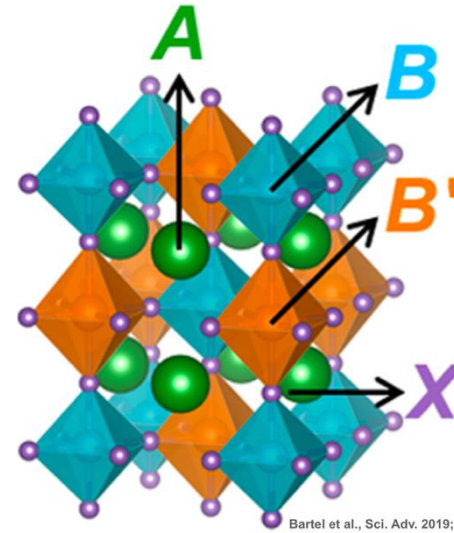
↓ for normal $y \mid x \sim N(\mu(x), \sigma^2(x))$

$$= (\mu(x) - y_s^*)(1 - \Phi(y_s^*)) + \sigma^2(x)\phi(y_s^*)$$

↑
“exploitation”

↑
“exploration”

Example: double Perovskite oxides (cubic)



Cations $X_A = \{\text{Li, Be, Na, Mg, ...}\}$

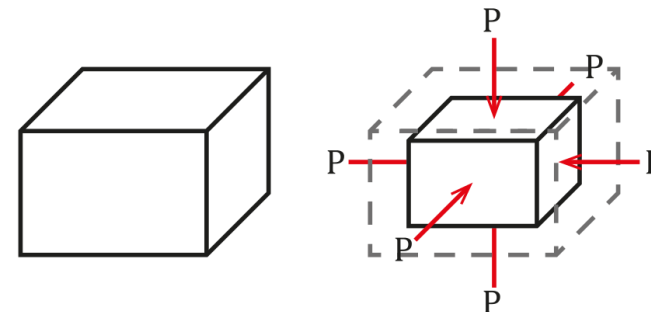
$X_B = \{\text{Li, Be, B, N, ...}\}$

$X_{B'} = \{\text{Be, B, C, N ...}\}$

Anion $X_X = \{\text{O}\}$

$$X = \{(a, b, b', x) \in X_A \times X_B \times X_{B'} \times X_X : b \leq b', \dots\}$$

$y(x) = -VdP/dV$: DFT computed bulk modulus



Gaussian Models yield Tractable Reward Estimation



Acquisition: expected improvement

$$\begin{aligned} u(x; s) &= \mathbb{E}(\max\{y - y_s^*, 0\}) \\ &= (\mu(x) - y_s^*)(1 - \Phi(y_s^*)) + \sigma^2(x)\phi(y_s^*) \end{aligned}$$

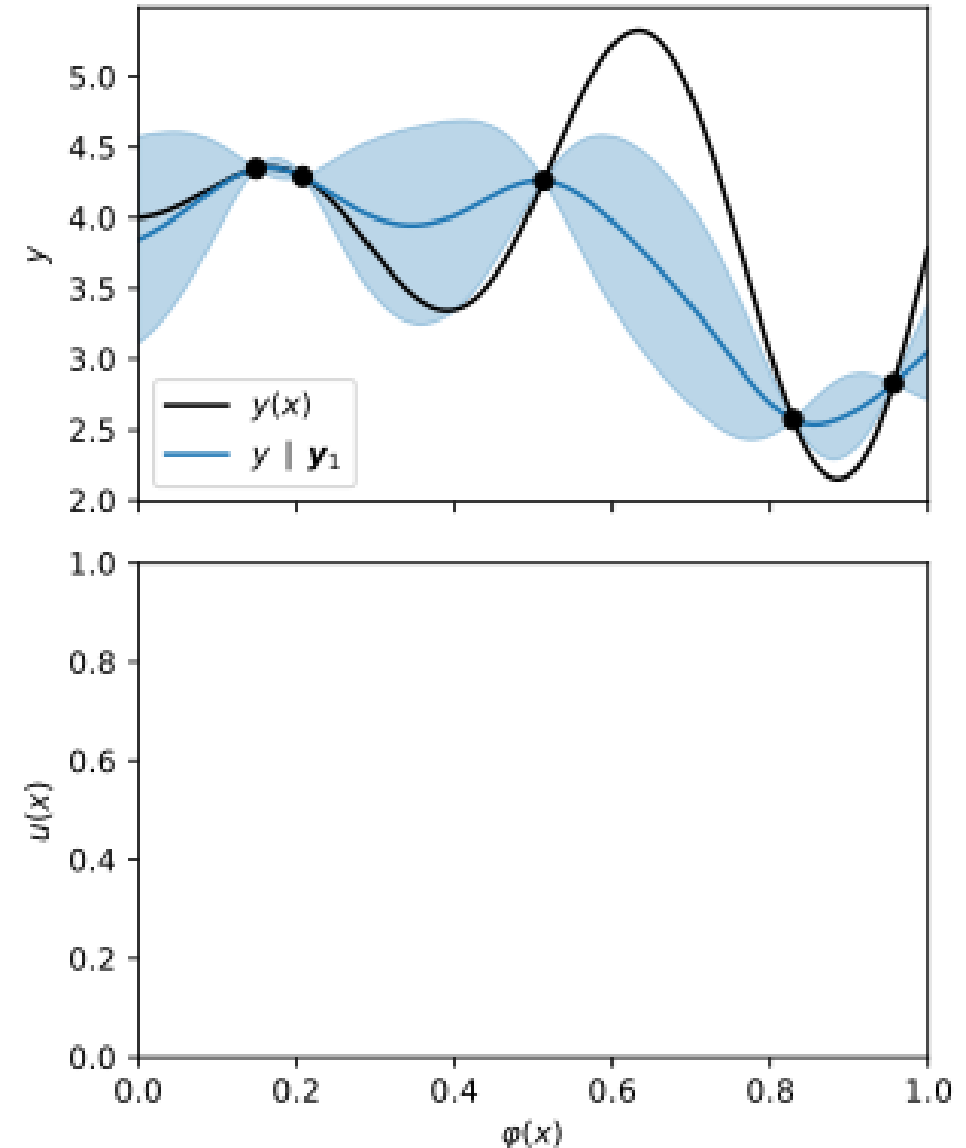
Gaussian process regression

$$\begin{aligned} \mathbf{y} &\sim N_{|X|}(\mathbf{0}, \mathbf{K} + \sigma^2 \mathbf{I}) \\ \mathbf{y}_2 | \mathbf{y}_1 &\sim N_{|X_2|}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{2,2}) \end{aligned}$$

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Random forest

$$\begin{aligned} f | \mathbf{y}_1, \mathbf{X}_1 &\sim \text{GreedyTree}(\tilde{\mathbf{y}}_1, \tilde{\mathbf{X}}_1) \\ y | \mathbf{y}_1 &\sim N(\mathbb{E}(f(\mathbf{x})), \mathbb{V}(f(\mathbf{x}))) \end{aligned}$$



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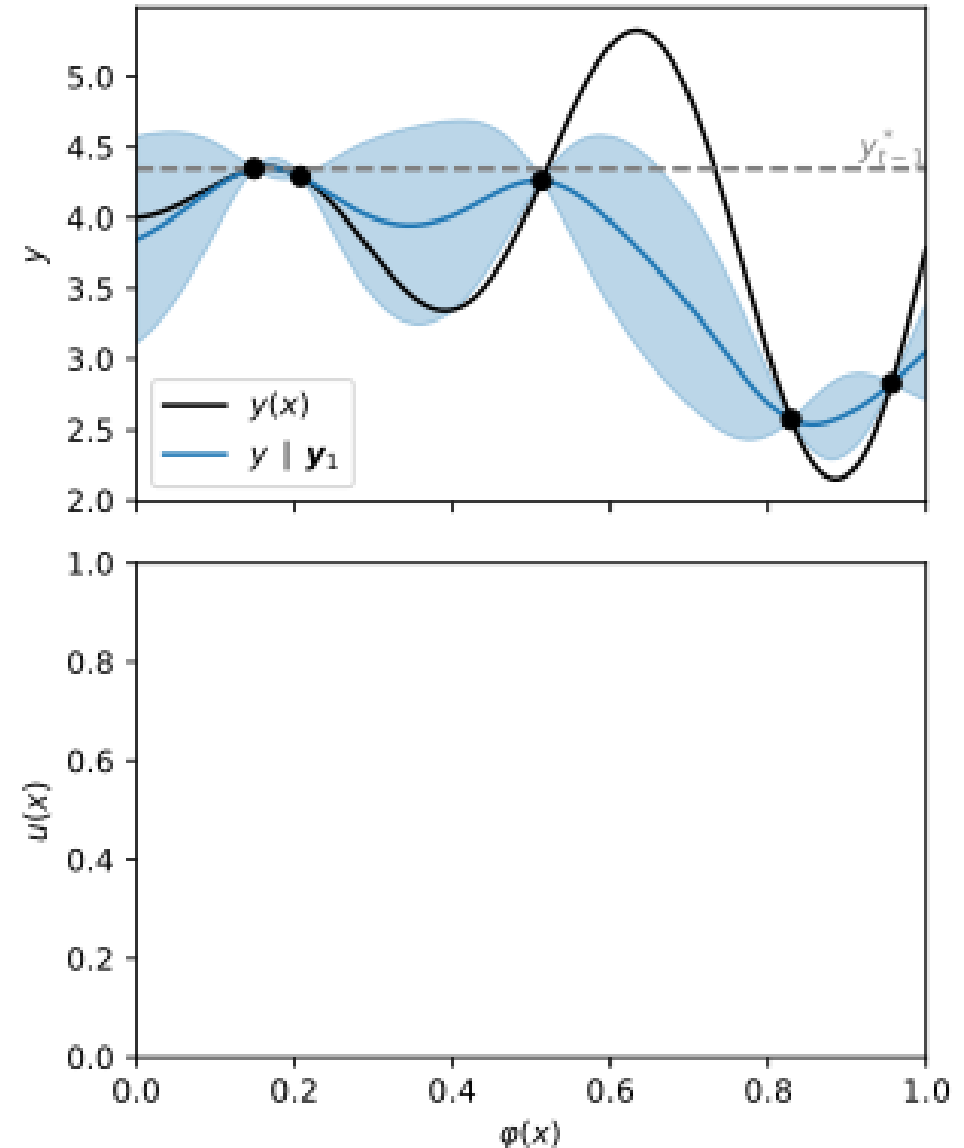
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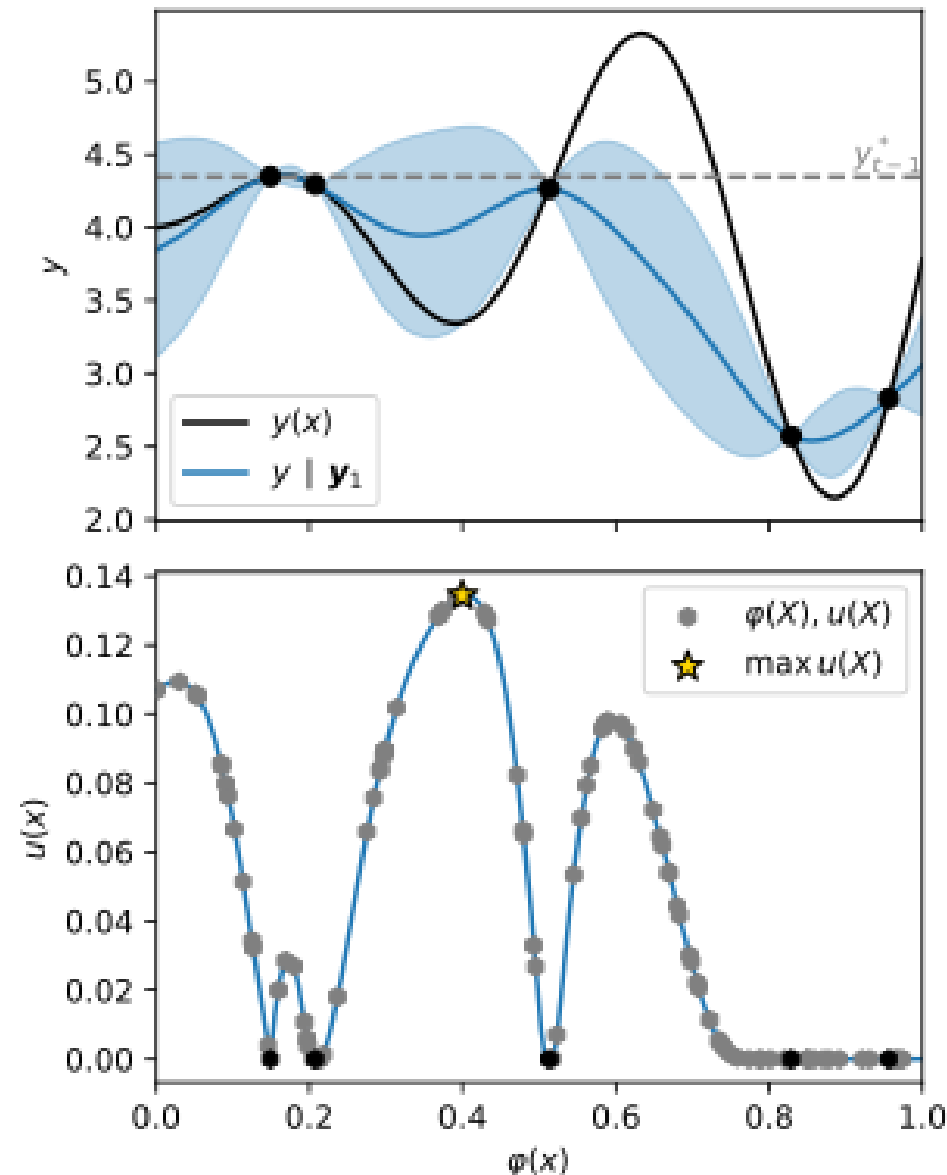
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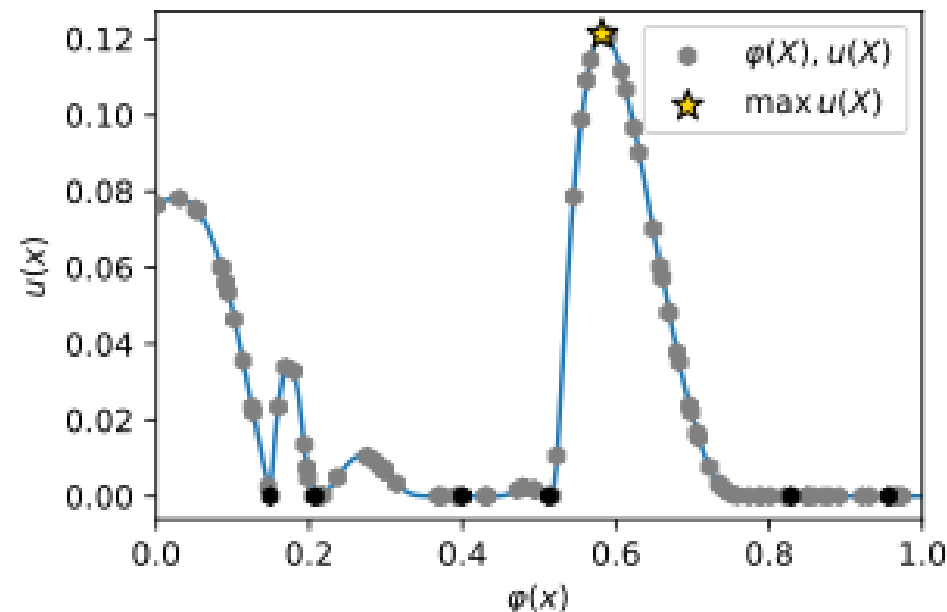
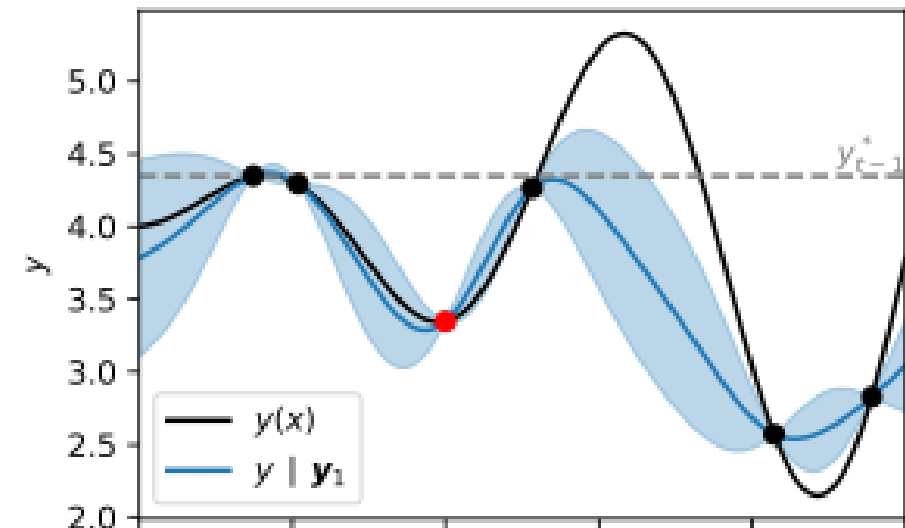
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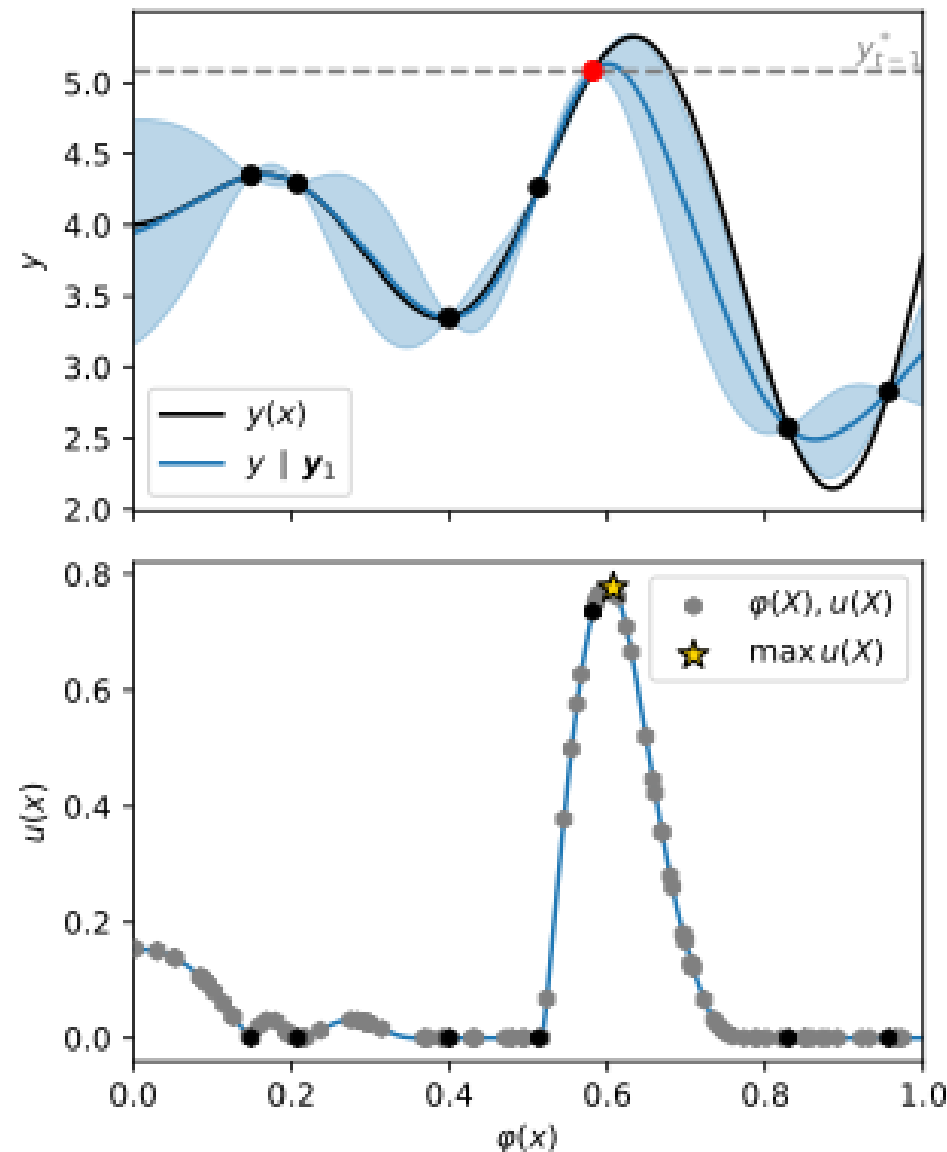
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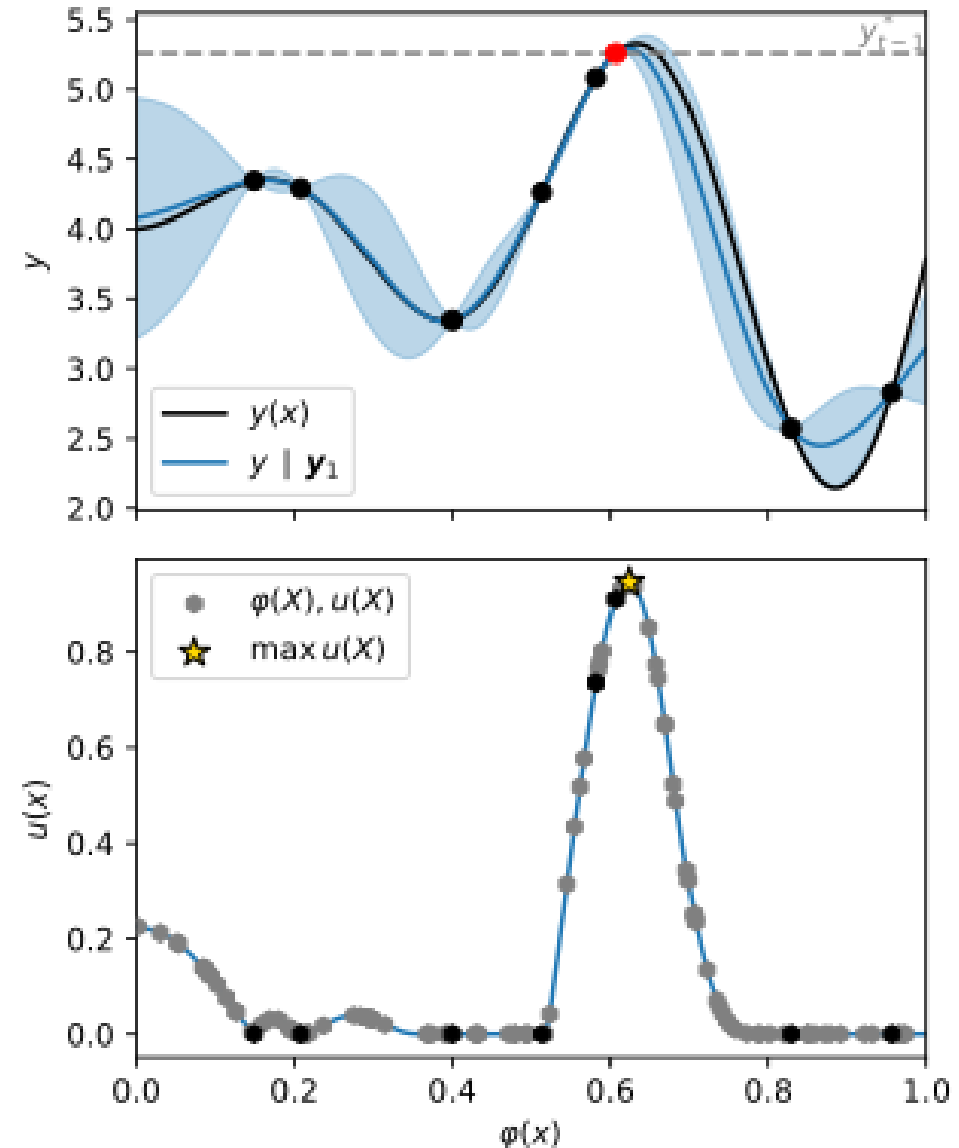
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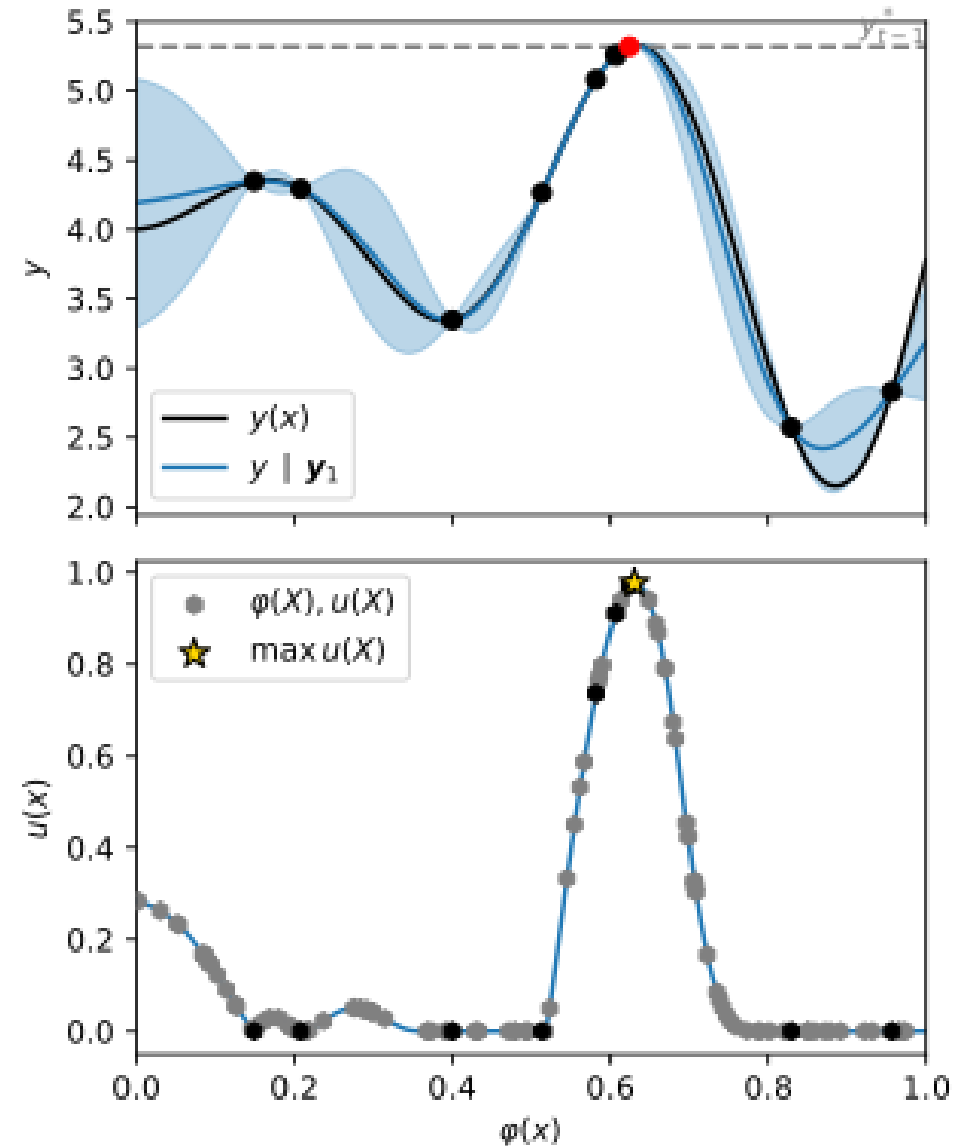
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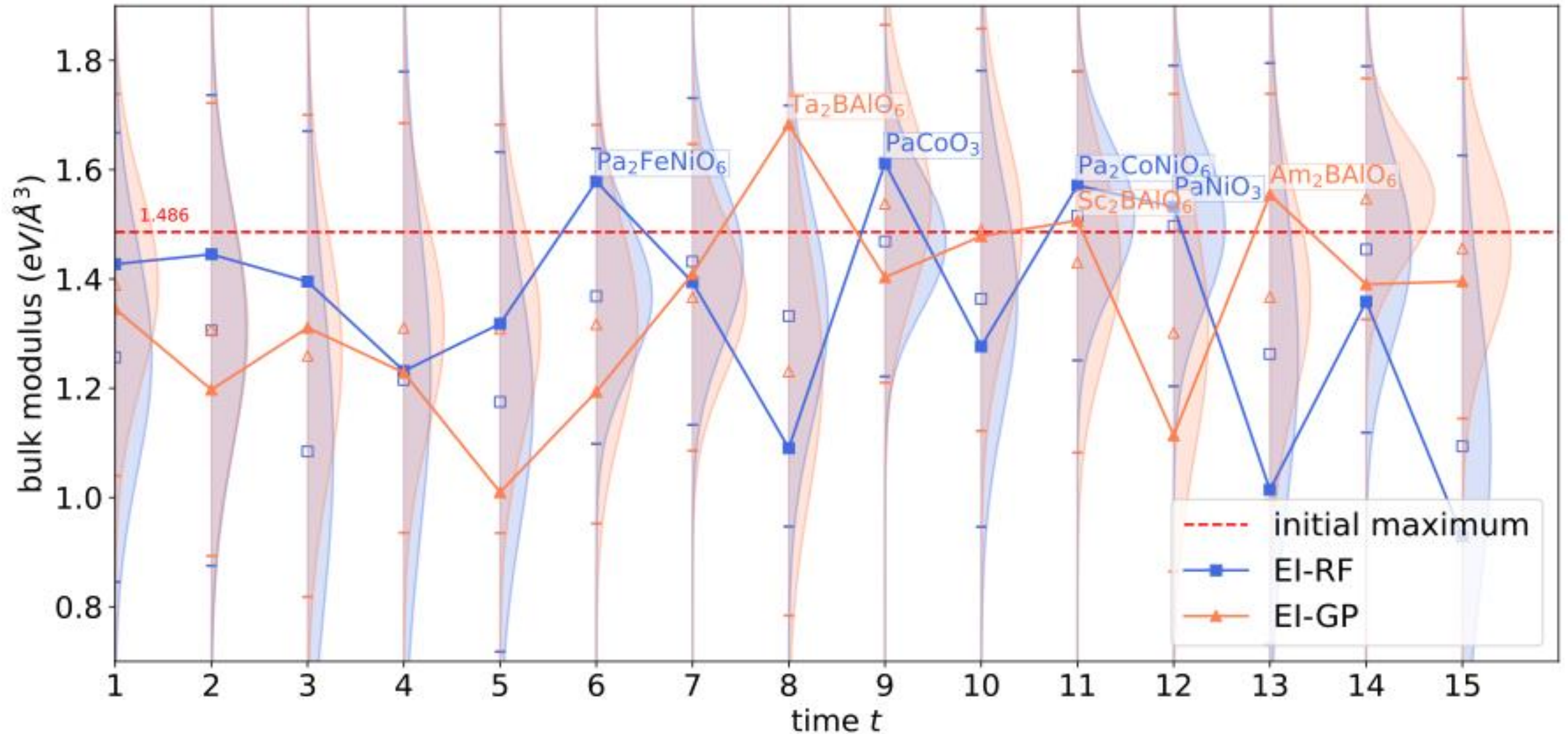
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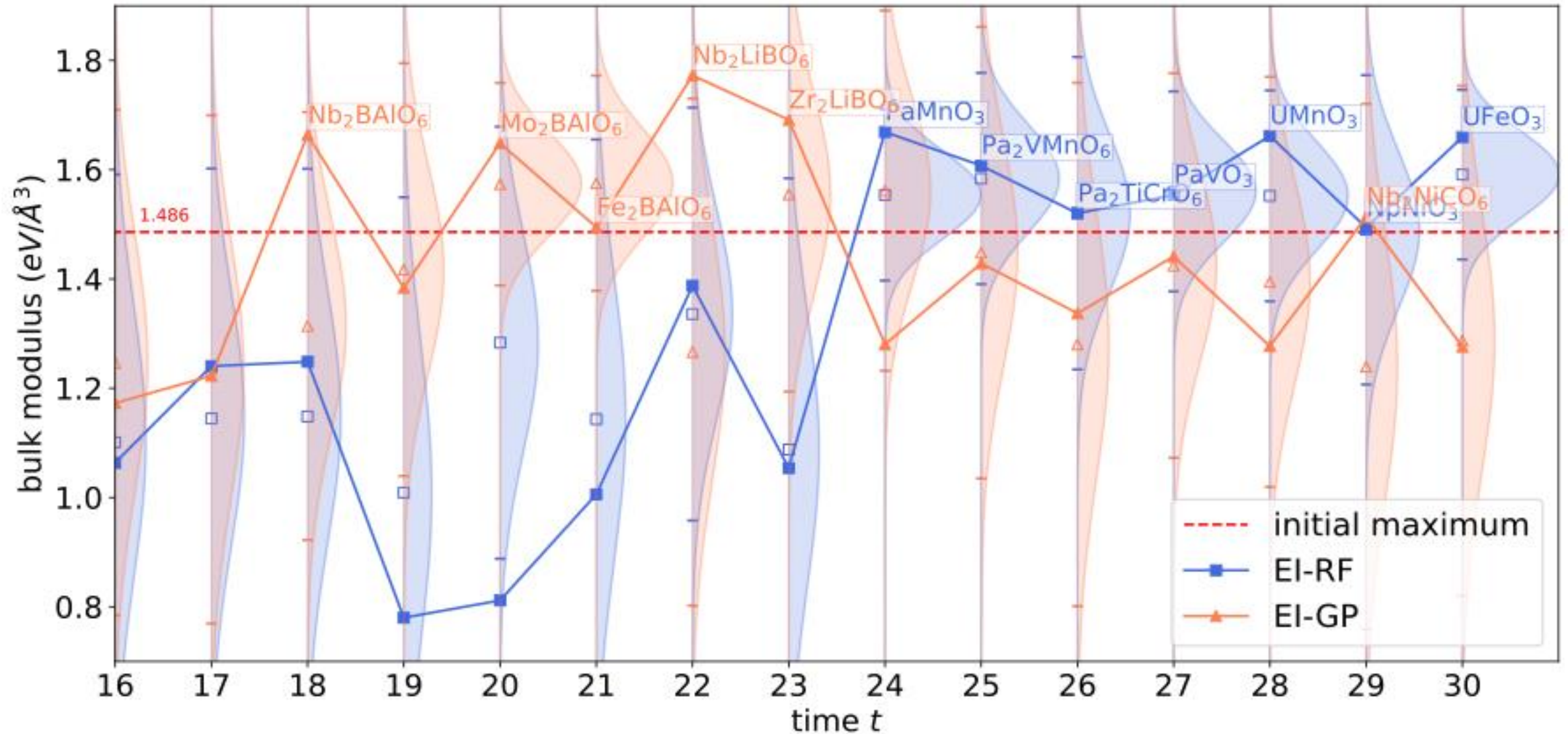
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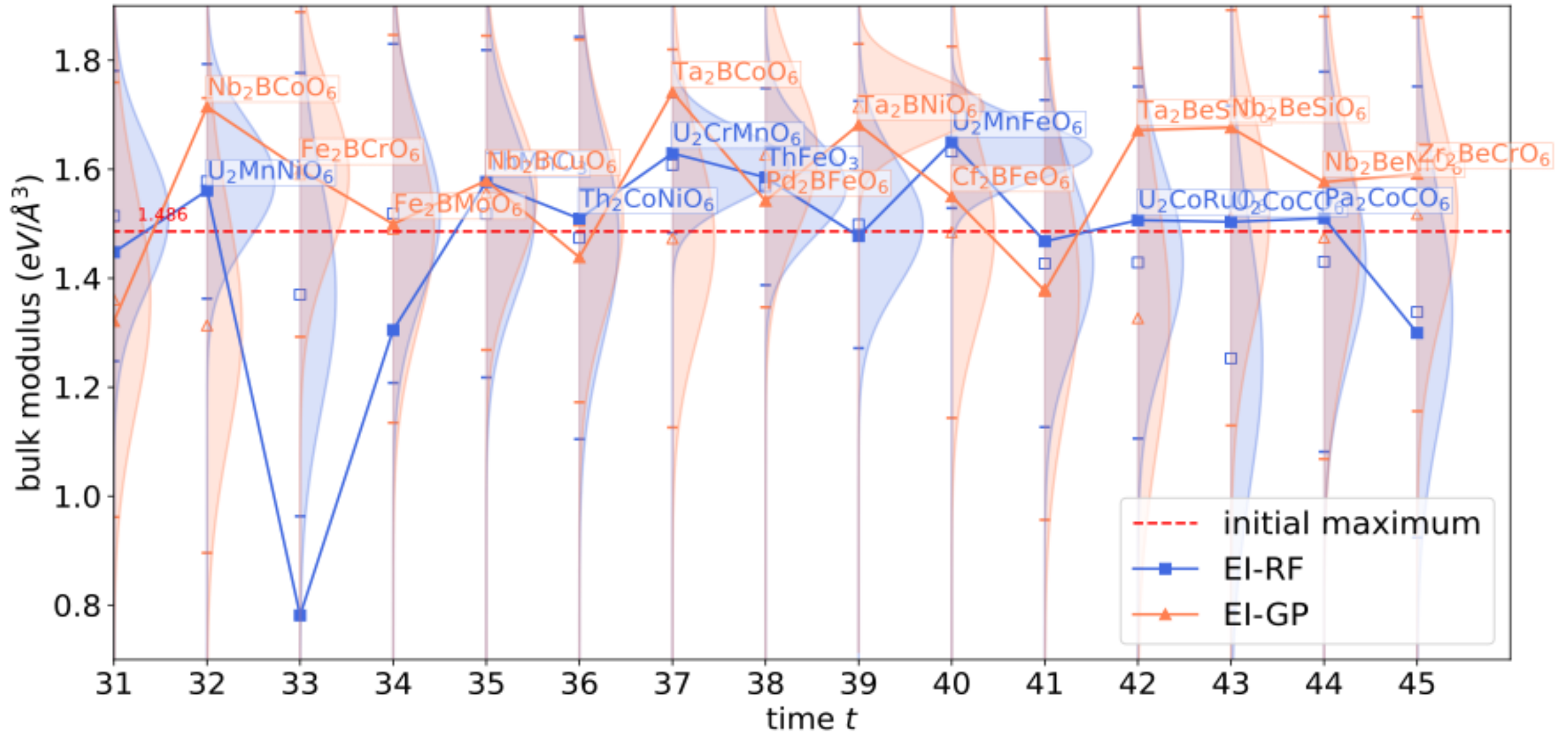
Strategy performs both: Exploration and Exploitation



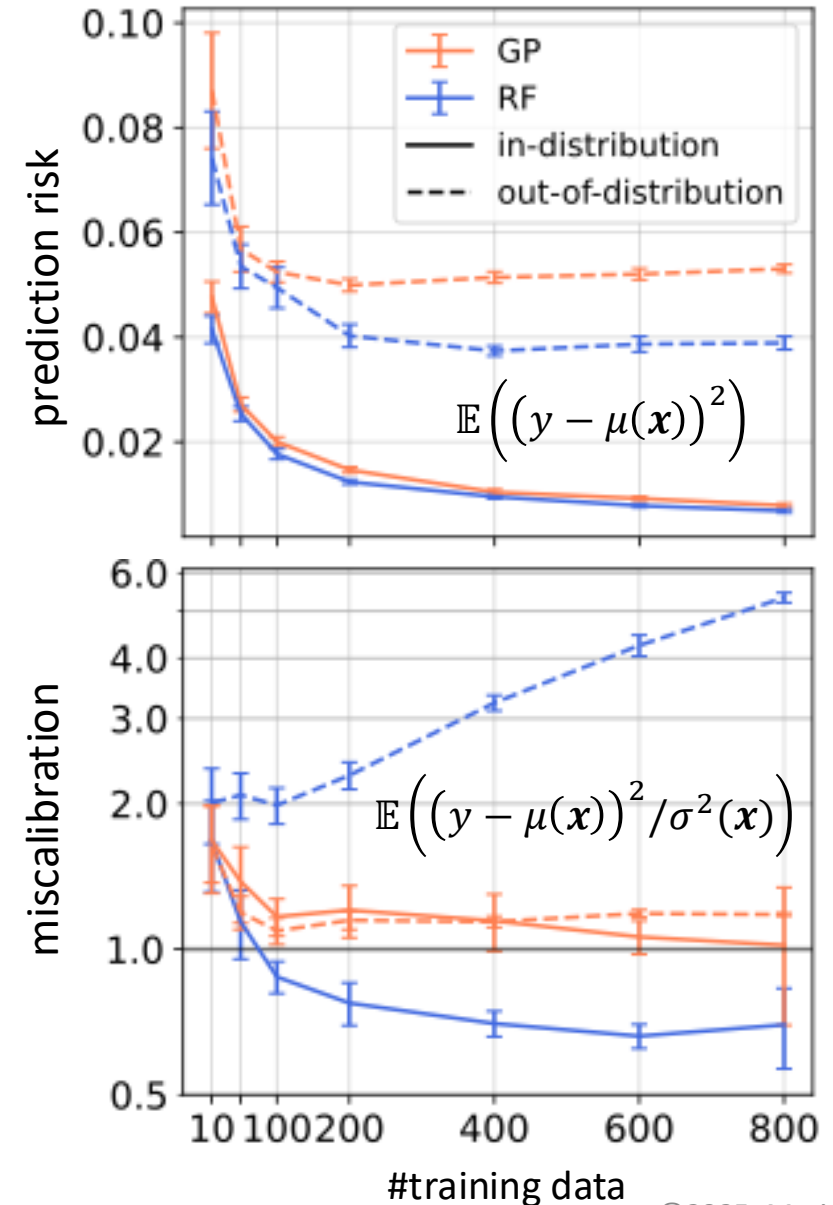
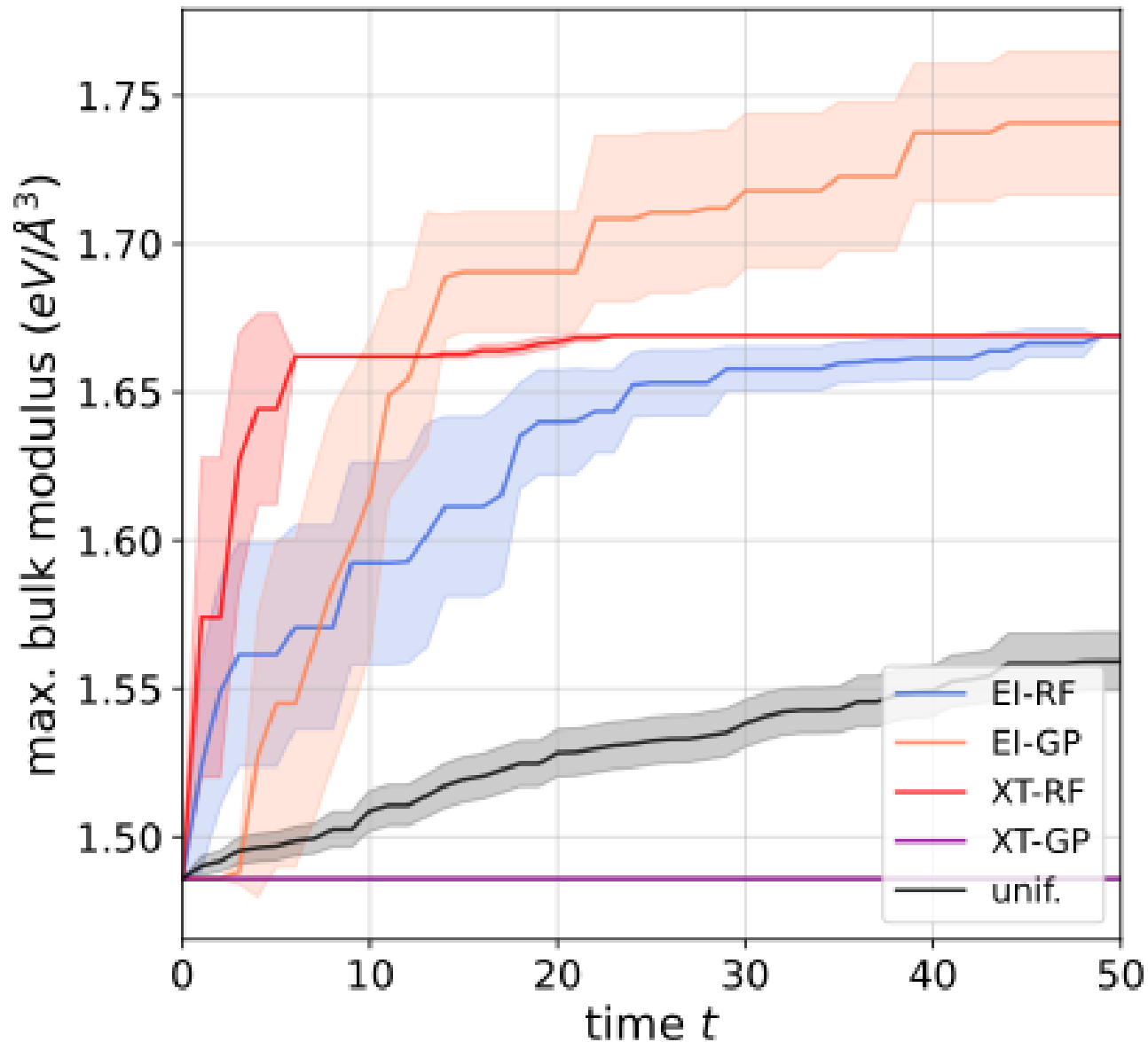
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Strategy performs both: Exploration and Exploitation



GP outperforms RF due to Uncertainty Quantification



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2. *Explorer*: mapping polymerisation-induced self-assembly phases

3. *Engineer*: optimising double perovskite hardness

4. *Collector*: discovering novel metal-organic frameworks

5. Problems and directions

Discovering Novel Metal-organic Frameworks



Phuc "Felix"
Luong



Zichao
Rong



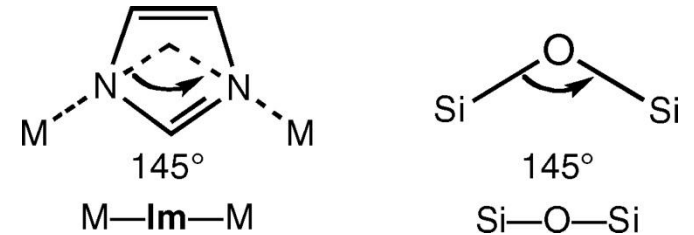
Omar
Yaghi



Zihao
Chen



Laurent
El Ghaoui



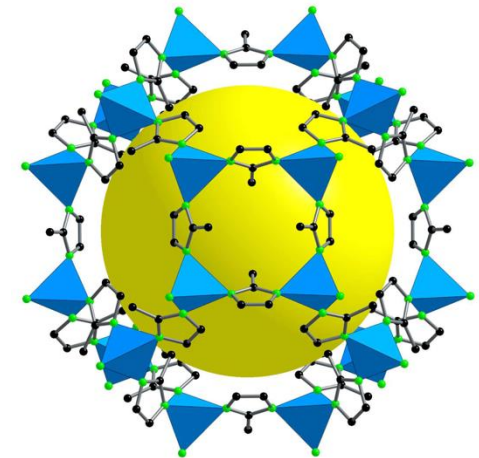
[Acc. Chem. Res. 2010, 43, 1, 58-67]



MONASH
University



VINUNIVERSITY



Goal: autonomous materials discovery platform

[Z Rong, Z Chen, P Luong et al. "Algorithmic iterative reticular synthesis of zeolitic imidazolate framework crystals." *Nature Synthesis* (2025): 1-11.]

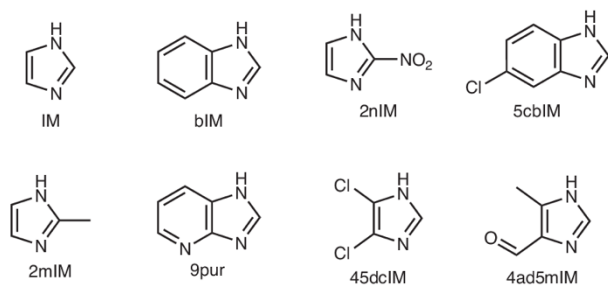
Collectors Maximise Number of Distinct Discoveries



Ideas

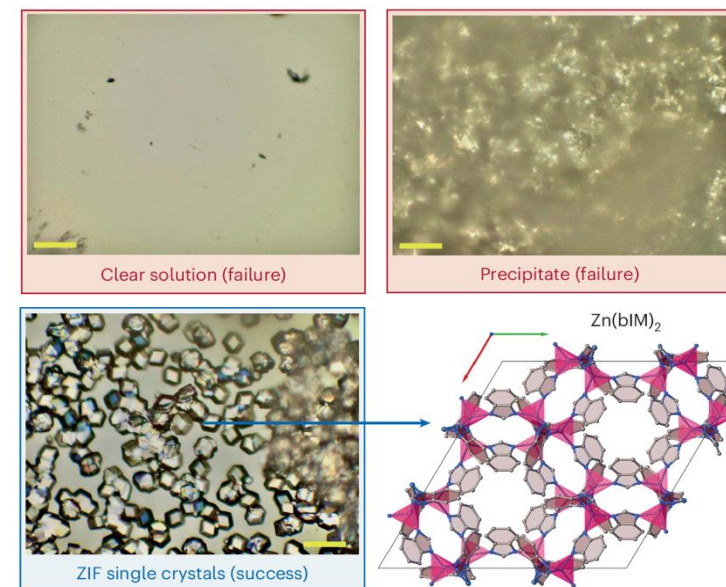
- Split inputs $x = (z, v)$ into “setup” z and “recipe” v
 - Choose desired **outcomes** $Y_+ \subset Y$
 - reward is **unique combinations** of setups and desired outcomes realized by a recipe
- $$r_t = \#\{(z, y) \in Z \times Y_+ : (z, v, y) \in s_t \setminus s_{t-1}\}$$

Example: ZIF synthesis



Z precursors,
e.g., $Z = \{\text{Im}, \text{bIM}, \text{2mIM}, \dots\}$

V experimental conditions,
e.g., $\text{Conc} \times \text{Temp} \times \text{Time}$



Y_+ targeted crystal structures
e.g., $Y_+ = \text{ZIF Crystals}$

Challenge: Experiment Batching



Single experiment expected reward:
unique count of new setup / succ. outcomes

$$\mathbb{E}(r \mid z, v) = \begin{cases} p(y \in Y_+ \mid z, v) & \text{if } \nexists v, (z, v, y) \in s \\ 0 & \text{otherwise} \end{cases}$$

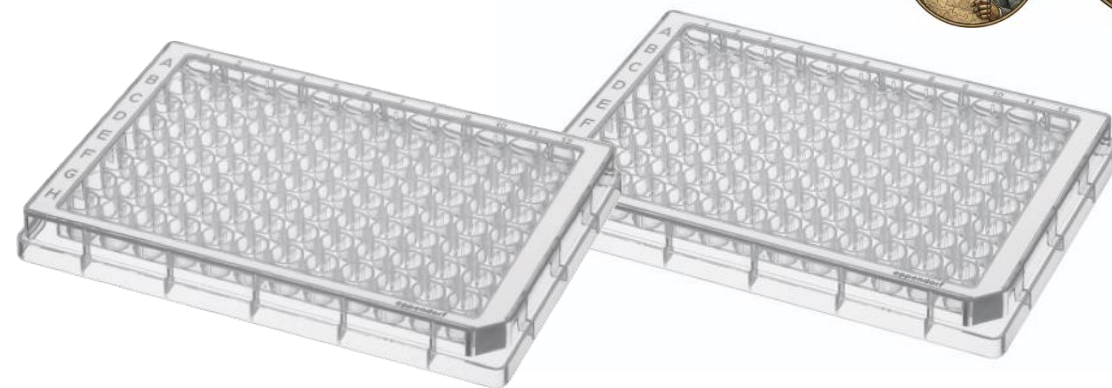
Batch actions $\mathbf{a} = (x_1, \dots, x_k) \in X^k$
 ...describe k experiments $x_i = (z_i, v_i)$

Expected batch reward:

$$\mathbb{E}(r \mid \mathbf{a}) = \sum_{(z, y) \notin s} \mathbb{P} \left(\bigcup_{i: z_i = z} \{y_i \neq y\} \mid \mathbf{a} \right)$$

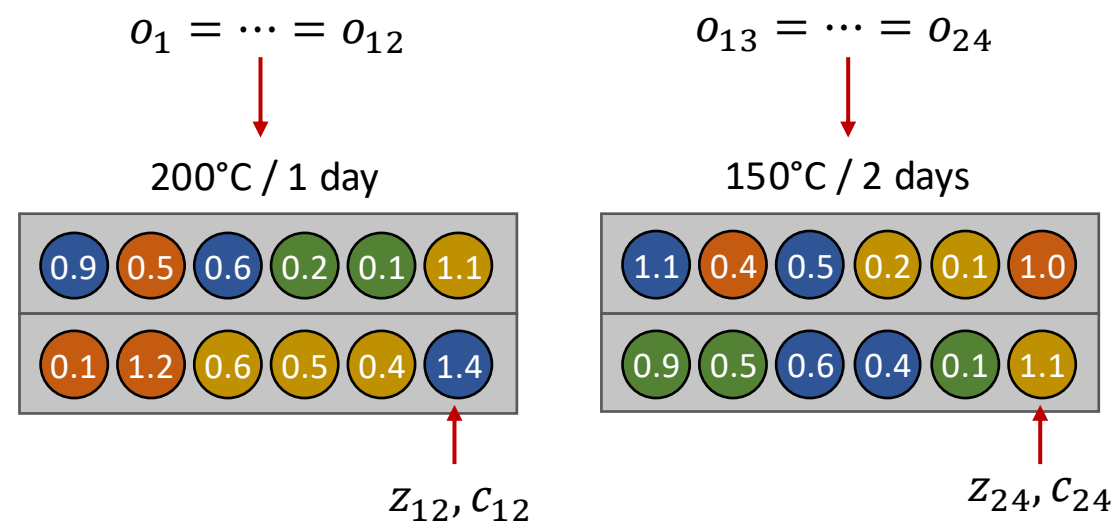
Constraint

- split recipe variables $v = (c, o)$
- only c can change per well / slot (*composition*)
- other vars o fixed per plate (*temp, time*)



microplates for high-throughput experimentation

Example: two 12-well plates



Need Algorithm for Complex Allocation Problem



Batch actions $\mathbf{a} = (x_1, \dots, x_k) \in X^k$

...describe s experiments $x_i = (z_i, v_i)$

Expected batch reward

$$\mathbb{E}(r \mid \mathbf{a}) = \sum_{(z,y) \notin s} \mathbb{P} \left(\bigcup_{i: z_i = z} \{y_i \neq y\} \mid \mathbf{a} \right)$$

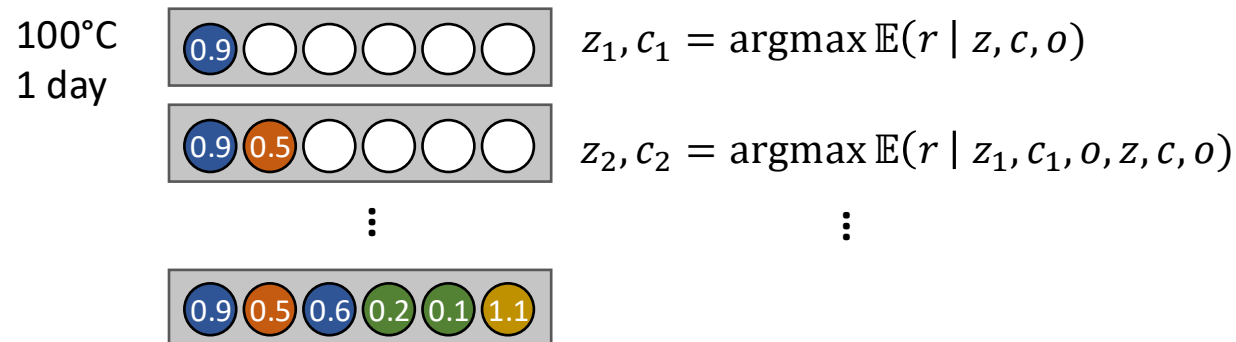


 precursor / outcome
joint success probability $p_{z,y}$

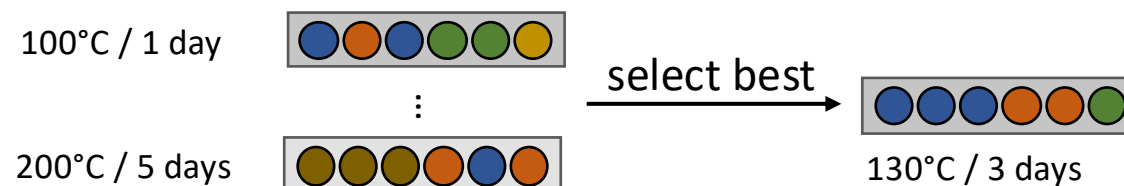
Constraint

- split recipe variables $v = (c, o)$
- only c can change per well / slot (*composition*)
- other vars o fixed per plate (*temp, time*)

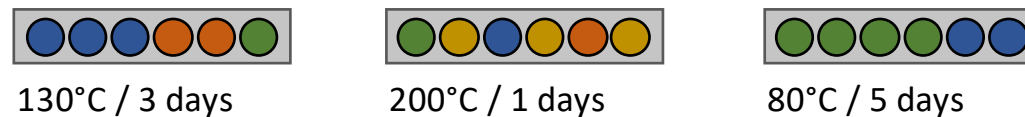
Greedy (z, c) -optimisation for fixed o



Brute-force condition enumeration



Greedy plate optimisation



Need to Estimate Joint Success Probabilities



Batch actions $\mathbf{a} = (x_1, \dots, x_k) \in X^k$

...describe s experiments $x_i = (z_i, u_i)$

Expected batch reward

$$\mathbb{E}(r \mid \mathbf{a}) = \sum_{(z,y) \notin s} \mathbb{P} \left(\bigcup_{i: z_i = z} \{y_i \neq y\} \mid \mathbf{a} \right)$$

precursor / outcome
joint success probability $p_{z,y}$

Can express success through failure probabilities

$$p_{z,y} = 1 - \prod_{i: z_i = z} q_{z,y}^{(i)}$$

$$q_{z,y}^{(i)} = \mathbb{P}\{y_i \neq y\} / \mathbb{P} \bigcap_{\substack{j: z_j = z \\ i < j}} \{y_j \neq y\}$$

Need to evaluate effect of experiment on batch reward

130°C / 3 days



200°C / 1 days



$\mathbb{E}(r \mid x_1, \dots, x_i, z, v)$



$\mathbb{E}(r \mid x_1, \dots, x_i, z', u')$

Idea: inject “hallucinated” failures

a_1	x_1	y_1
	...	...
	x_k	y_k
a_2	x_{k+1}	y_{k+1}
	...	...
	x_{2k}	y_{2k}
	x_{2k+1}	0
	x_{2k+2}	0

state s_2

injected points

$$q_{z,y}^{(i)} = p(y \mid z; s_t \cup \{(x_j, 0) : z_j = z, j < i\})$$

Design Models based on Linker Representation

Random forest classifier

$$y \mid \mathbf{x} \sim \text{Bern}\left(\mathbb{E}(f(\varphi(\mathbf{x})))\right)$$

$$f \mid \mathbf{y}_1 \sim \text{GreedyTree}(\tilde{\mathbf{y}}_1, \tilde{\mathbf{X}}_1)$$

Gaussian process classifier

$$\mathbf{f} \sim \mathcal{N}_{|X|}(\mathbf{0}, \mathbf{K}) \xrightarrow{\text{red arrow}} k_{i,j} = \exp\left(-\|\varphi(x_i) - \varphi(x_j)\|^2 / 2\alpha\right)$$

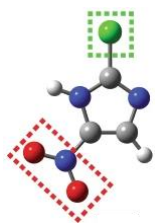
$$y \mid f \sim \text{Bern}(\sigma(f))$$

$$\mathbf{f}_2 \mid \mathbf{y}_1 \sim \mathcal{N}_{|X_2|}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{2,2})$$

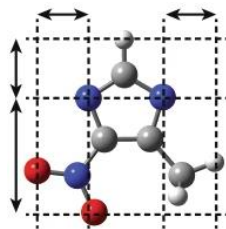
$$\mathbf{K}_{2,1} \mathbf{K}_{1,1}^{-1} \hat{\mathbf{f}}_1$$

$$\mathbf{K}_{2,2} - \mathbf{K}_{2,1}(\mathbf{K}_{1,1} + \mathbf{W}^{-1})^{-1} \mathbf{K}_{1,2}$$

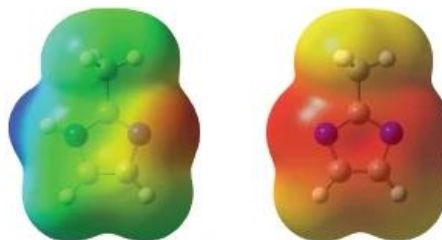
Features:



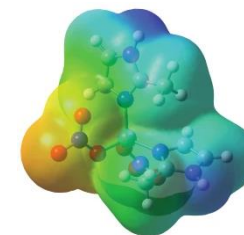
Functional group counts



Linker dimensions

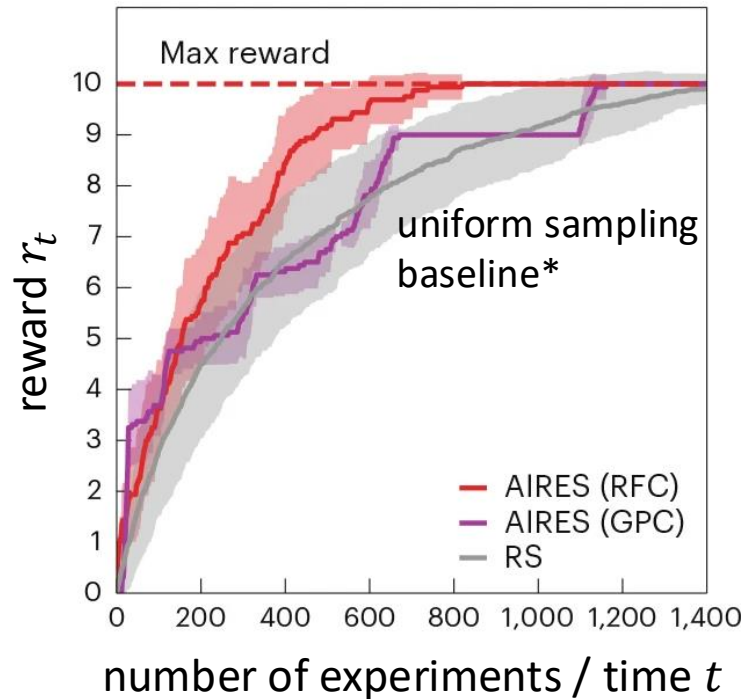


QM properties of neutral/deprotonated linker

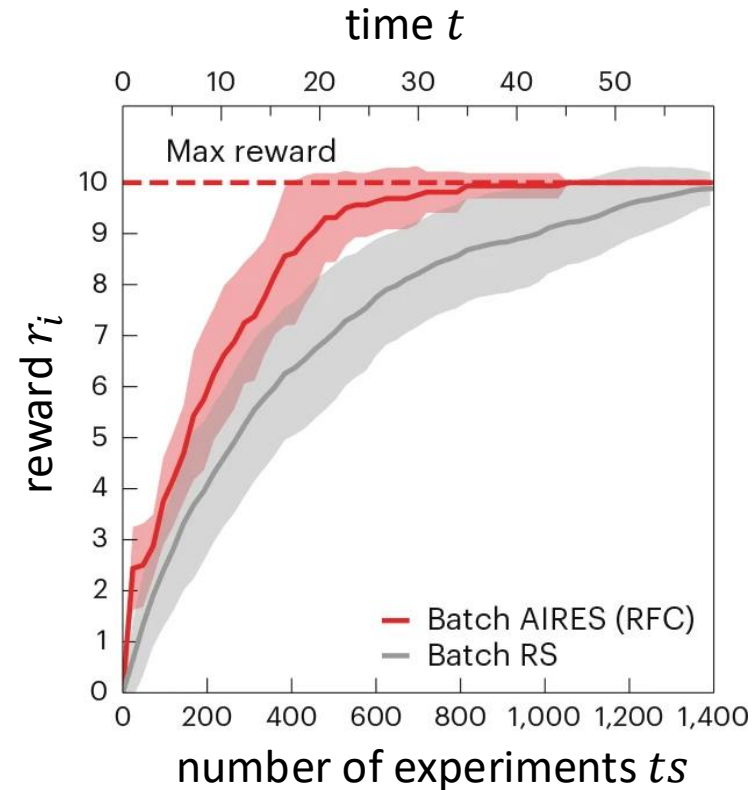


QM properties of pre-nucleation species

RFC-Collector retains Advantage in Batch Mode



Sequential



Batch (2 plates, 12 wells each)

$$*a_t \sim \text{Unif}\{(z, v) : \exists y \in Y_+, \nexists v', (z, v', y) \in s_t\}$$

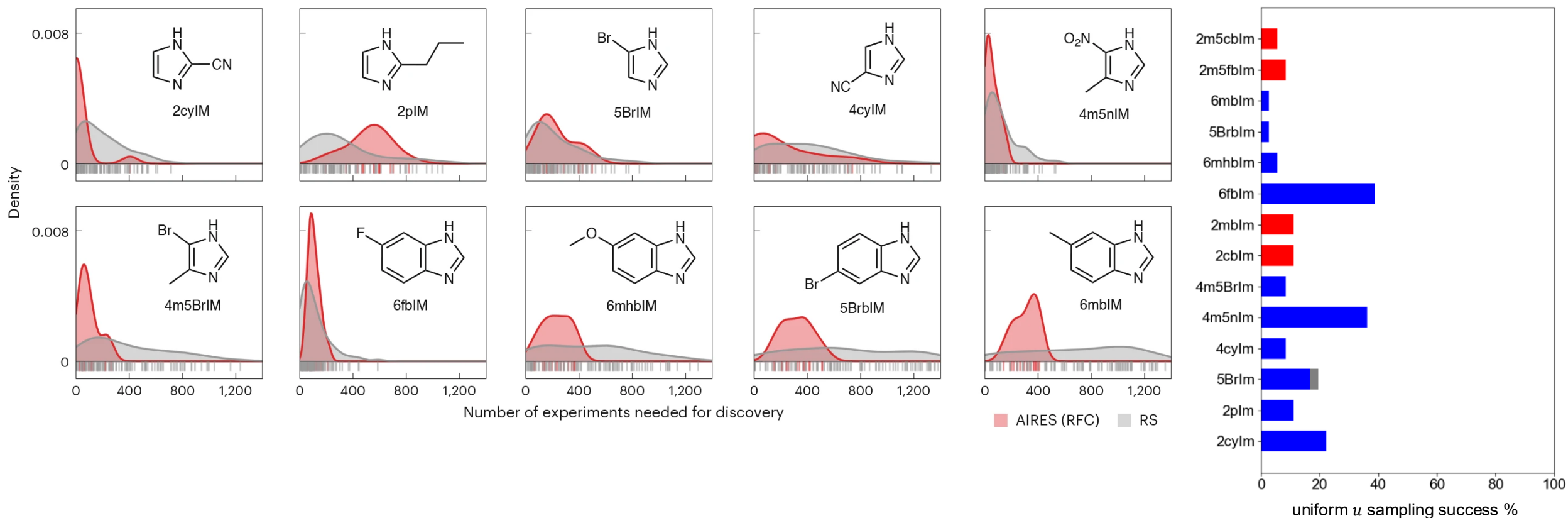
Strategy Drastically Shrinks Tails of Discovery Times

Time to discovery of recipe leading to outcome $y \in Y_+$ for setup z :

$$t(z, y) = \min\{t \in \mathbb{N} : \exists v, (z, v, y) \in s_t\}$$

Depends on required recipe specificity:

$$\mathbb{E}_{v \sim \text{Unif}(V)} p(y_+ | z, v)$$



Outline

1. Personas of Agentic AI

2. *Explorer*: mapping polymerisation-induced self-assembly phases

3. *Engineer*: optimising double perovskite hardness

4. *Collector*: discovering novel metal-organic frameworks

5. Problems and directions

Directions to Explore in Collaborations

1. Interpretability: rule-based models
2. Representation Learning: GNNs, LLMs, pre-training
3. Uncertainty quantification
4. Grid-free optimisation
5. Experiment look-ahead and batching